

# RIME Computational Companion Archive

*Versioned Reproducibility Data, Computational Observations, Open Problems, and Historical Records*

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**Archive boundary.** This archive is optional human-readable research companion material. Papers I, II, and III are mathematically self-contained and do not rely on it as a premise, definition source, executable certificate, or claim authority. It is not a paper, theorem source, semantic authority, or prerequisite for those papers. Mathematical claims must be cited from the corresponding papers. The archive preserves reproducibility data, computational observations, open conjectures, and versioned historical records with explicit status labels. Executable certificates are controlled by their declared scripts and structured artifacts; corrections affecting interpretation are recorded in `HISTORY.md`.

## Independent Paper Navigation

The entries below are navigation records, not a dependency chain. Papers I, II, and III have independent Zenodo records.

| Paper | Independent title                                                                                                                          | Repository                                                                | Separate Zenodo DOI                                                           |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| I     | <i>Spectral Sector Decomposition in the Rubik's Cube Representation: Block Spectral Structure and a Conditional Rationality Criterion</i>  | <a href="#">PDF</a> , <a href="#">source</a> , BibTeX <code>paper1</code> | <a href="https://doi.org/10.5281/zenodo.21571403">10.5281/zenodo.21571403</a> |
| II    | <i>Noncommutative Transport Topology in the Rubik's Cube Representation: Sector Non-Invariance, Direct Support, and Transport Channels</i> | <a href="#">PDF</a> , <a href="#">source</a> , BibTeX <code>paper2</code> | <a href="https://doi.org/10.5281/zenodo.21581072">10.5281/zenodo.21581072</a> |
| III   | <i>Support-Graph Reachability and Matrix-Composition Obstructions: Image-Kernel Mismatch with a Rubik-Cube Case Study</i>                  | <a href="#">PDF</a> , <a href="#">source</a> , BibTeX <code>paper3</code> | <a href="https://doi.org/10.5281/zenodo.21583070">10.5281/zenodo.21583070</a> |

The immutable combined-release DOI [10.5281/zenodo.21108197](https://doi.org/10.5281/zenodo.21108197) remains historical provenance. It does not define the current Paper I–III architecture and must not be reused as the CCS v2.1 DOI. That historical combined package is a provenance record only.

**Paper II v2 routing.** The Paper II theorem spine is the exact Transport–Non-Invariance Identity and direct block-locality theorem. The nine-sector registration,

ten-edge graph, Type I/II labels, and EP algebra census are computational certificates. Block-level  $\text{Supp}_{\text{nc}}$  is the family-level maximum over all three per-axis QT commutator pairs and is only a candidate localizer: its 15 overlap pairs contain the nine Type I labelled edges and six nonedges. It is not a sufficient transport criterion. Generator-family field tables, S3 negative controls, full EP algebra tables, and auxiliary figures remain CCS material and are not manuscript premises.

## Reproducibility Architecture

The repository uses four separated layers:

| Layer | Name                 | Files                                       | Role     | Rule                                                                               |
|-------|----------------------|---------------------------------------------|----------|------------------------------------------------------------------------------------|
| 1     | Papers               | <code>papers/paper*/</code>                 | Claim    | Self-contained theorem, certificate, observation, and research-program boundaries. |
| 2     | Executable artifacts | <code>experiments/paper*/, results/</code>  | Certify  | Declared scripts, parameters, hashes, residuals, and structured outputs.           |
| 3     | CCS v2.1             | <code>ccs/canonical_specification.md</code> | Review   | Human-readable extended data, observations, open questions, and selected history.  |
| 4     | Private raw archive  | not distributed                             | Preserve | Giant matrices, abandoned runs, and exploratory provenance.                        |

CCS v2.1 is not a copy of the raw archive and does not replace executable artifacts. It is a curated review layer. The papers state their claims, the declared scripts and structured artifacts certify finite computations, and `HISTORY.md` records corrections that affect interpretation.

## Navigation and Citation Boundary

Papers I–III do not cite CCS v2.1 as a scholarly or mathematical authority. Internal part, table, and figure labels below are navigation aids for this archive only. Paper III numerical claims are defined by its own manuscript and matrix certificate.

**Internal part references.** Archive-local links use Roman numeral prefixes:

| Prefix         | Scope                                                      | Example                |
|----------------|------------------------------------------------------------|------------------------|
| CCS Part 0     | Global Reference Map (notation tables, terminology)        | (CCS Part 0)           |
| CCS Part 0.5   | Canonical API Surface                                      | (CCS Part 0.5)         |
| CCS-I          | Part I — Core Numerical Structures (§1, §2)                | (CCS-I §2.1)           |
| CCS-II         | Part II — Extended Computational Observations (§II.1–II.4) | (CCS-II §II.4)         |
| CCS-III        | Historical derivation index; excluded from the release PDF | source provenance only |
| CCS Appendix X | Appendices A–F                                             | (CCS Appendix F)       |

**Section numbering.** Part I uses §1.x (spectral objects) and §2.x (numerical data). Part II uses §II.1–II.4 for extended observations. The historical Part III numbering is retained in source provenance only. Appendix subsections use letter prefixes (§A.1, §B.1, §E.1, §F.1).

**Tables and figures.** (CCS Table C3), (CCS Fig. C1) — figures are captioned where they appear. Historical images are repository provenance and are not separately indexed in this release.

The Terminology Convention at the end of Part 0 defines the four canonical terms: QT/HT joint-spectral sector, hybrid sector, transport-active, canonical sectorization.

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## Archive Status Tags

Every retained item must carry one of the following statuses in context:

| Status                            | Meaning                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Theorem / exact derivation</b> | Finite mathematical statement proved under explicit hypotheses; the independent paper remains the preferred citation. |
| <b>Computational certificate</b>  | Declared realization, dtype, tolerance, algorithm, artifact, and reproducible script are available.                   |
| <b>Computational observation</b>  | Finite pattern or numerical recognition without a full promotion certificate.                                         |
| <b>Research program</b>           | Conjecture, proposed hierarchy, genericity question, or future experiment.                                            |
| <b>Historical / withdrawn</b>     | Provenance only; excluded from the PDF or displayed with an explicit historical warning.                              |

## Box Conventions

Callout boxes separate exact statements, registered data, warnings, and provisional findings. Their labels describe archive status; they do not make CCS v2.1 an independent claim authority:

| Box                                             | Style                                       | Purpose                                                                                                         |
|-------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Theorem / Lemma / Corollary / Definition</b> | Blockquote > with bold label                | Formal mathematical statement. Proofs appear inside the box, set off with <i>Proof.</i> or <i>Proof sketch.</i> |
| <b>Registered</b>                               | Blockquote > with <b>Registered.</b> label  | Finite data tied to a declared computational realization.                                                       |
| <b>Warning</b>                                  | Blockquote > with <b>Warning.</b> label     | Important constraint, pitfall, or normative requirement (SHALL/MUST). Non-negotiable.                           |
| <b>Exploratory</b>                              | Blockquote > with <b>Exploratory.</b> label | Observation, conjecture, or research-program item subject to revision.                                          |

## 1 Executive Guide

This guide provides the shortest reliable reading path through the archive. The subsequent sections preserve tables, implementation context, figures, failed candidates, and research history. They do not enlarge the claims of the independent papers.

## Current Finite Records

| Record                                                                          | Current status                                                                          | Owning source                                            |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------|
| Six displayed averaging layers with dimension census (20,2,39,26,106,35)        | Computational observation; exact and conditional statements are separated in Paper I    | Paper I and <code>experiments/paper1/validation/</code>  |
| Nine registered QT/HT joint-spectral sectors                                    | Numerical registration on the declared complex128 realization                           | Paper II and <code>experiments/paper2/validation/</code> |
| Symmetric ten-edge direct graph with degree sequence (0,2,2,2,2,5,3,1,3)        | Computational certificate                                                               | Paper II                                                 |
| Fifteen noncommutative-support candidates: nine Type I edges and six nonedges   | Computational certificate; the localizer is not sufficient                              | Paper II                                                 |
| EP algebra census: four $M_2(\mathbb{C})$ components and four scalar components | Computational certificate supported by the finite-dimensional unital *-algebra argument | Paper II                                                 |
| Five support-graph paths whose evaluated projected products are machine-zero    | Computational certificate with image-kernel/block obstruction interpretation            | Paper III                                                |

These five declared two-step support-graph paths form the finite Paper III composition-obstruction audit.

## Promotion Boundaries

- Numerical QT/HT commutation and clustering do not prove an exact labelled joint spectral resolution.
- Numerical recognition of rational or quadratic values does not determine an exact spectral field.
- A direct-support path does not guarantee a nonzero routed matrix product.
- A routed product does not automatically determine a full word, commutator, or Lie depth.
- Ambient incidence codimension is a benchmark, not the codimension of a representation-derived pullback.

## Research Seeds Preserved Here

The archive retains three concrete promotion targets: exact QH algebra registration; exact characteristic/minimal-polynomial certificates for generator-family arithmetic contrasts; and structured pullback geometry for representation-derived incidence. Earlier T7, commutant-restriction, completion, search, and spectral-triple narratives are historical records only.

## Part 0 — Global Reference Map

**Purpose.** Human-readable lookup for registered Paper I–II numerical objects, implementation locations, and shared Rubik notation. The independent papers define their own mathematical objects and claims locally.

**How to use.** For any symbol or concept name, use column 3 to locate the paper definition or the archive record and column 4 to identify its scope. Paper definitions control mathematical meaning; archive entries provide data and provenance only.

**Registered default.** Unless otherwise stated, the archived nine-sector decomposition is the numerical joint-spectral registration associated with

$$\mathcal{B}_{\text{QH}} = \text{alg}(A_{18}, \text{QT}_{\text{all}}, \text{HT}_{\text{all}}) = \text{alg}(\text{QT}_{\text{all}}, \text{HT}_{\text{all}}),$$

where  $A_{18} = (2/3)\text{QT}_{\text{all}} + (1/3)\text{HT}_{\text{all}}$ . The displayed algebraic interpretation is conditional on exact commuting-Hermitian registration. Sectorizations involving auxiliary block projectors (e.g.  $P_{\text{nat}}$ ) are separate declared realizations.

## Layer A — Static Spectral Structure (Paper I)

Core object:  $A = \frac{1}{|S|} \sum_{g \in S} \rho(g)$

| Symbol                                                                                  | Concept                                                                                                                 | First Defined         | Used In            |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------|
| $\rho : G \rightarrow \text{GL}(228, \mathbb{C})$                                       | Rubik's cube representation                                                                                             | Paper I §2            | I, II, III         |
| $V =$<br>$V_{\text{cp}} \oplus V_{\text{ep}} \oplus V_{\text{co}} \oplus V_{\text{eo}}$ | Block decomposition ( $64 + 144 + 8 + 12 = 228$ )                                                                       | Paper I §3.4          | I, II, III         |
| cp, ep, co, eo                                                                          | Four invariant blocks                                                                                                   | Paper I §2            | I, II, III         |
| $A = \frac{1}{ S } \sum_{s \in S} \rho(s)$                                              | Averaging operator                                                                                                      | Paper I §2            | I, II, III         |
| $V_{\lambda}, \lambda = 1 - k/9$                                                        | Canonical layers (6), eigenvalue form                                                                                   | Paper I §3            | I, II, III         |
| $k \in \{0, 1, 2, 3, 4, 6\}$                                                            | Admissible $k$ -set (6 values, $k = 5$ vacant)                                                                          | Paper I §3,<br>§7.3   | I, II, III         |
| $P_{\lambda}^A$                                                                         | Orthogonal projector onto the $A$ -eigenspace $V_{\lambda}^A$                                                           | Paper I §3.1          | I, II              |
| $\mathcal{K}(A) = \bigcup_B \mathcal{K}_B$                                              | Blockwise $k$ -set union formula                                                                                        | Paper I §7.3          | I                  |
| $\chi_{\lambda}(s) =$<br>$\text{Tr}(P_{\lambda} \rho(s))$                               | Eigenspace trace                                                                                                        | Paper I §3.1          | I                  |
| $\omega + \omega^2 + 1 = 0$                                                             | $\mathbb{Z}_3$ phase cancellation                                                                                       | Paper I §4.1,<br>§7.1 | I                  |
| $h_i =$<br>$\frac{1}{2}(\rho(g_i) + \rho(g_i^{-1}))$                                    | Per-generator Hermitian average                                                                                         | Paper I §7.2          | I                  |
| $V = \bigoplus_{\mu} V^{(\mu)}, C_{\mu}$                                                | Ambient $G$ -isotypic decomposition and its central projectors; distinct from the $A$ -spectral decomposition           | Paper I App<br>B      | I                  |
| $\text{Tr}(P_{\lambda}^A C_{\mu} P_{\lambda}^A)$                                        | Ambient-isotypic overlap mass; not a subrepresentation multiplicity unless the projectors commute                       | Paper I App<br>B      | I                  |
| $O, U_R$                                                                                | Registered orientation-preserving cubic rotation action commuting numerically with $A$ ; distinct from transport by $G$ | Paper I §4,<br>App B  | I                  |
| $\dim \text{End}_G(V) =$<br>610                                                         | Candidate ambient-commutant dimension; unpromoted pending exact certificate                                             | CCS legacy<br>§2.8    | provenance<br>only |

## Layer B — Discrete Transport Structure (Paper II)

Core object:  $K_{\beta\alpha}^S = \max_{s \in S} \|Q_{\beta} \rho(s) Q_{\alpha}\|_F$

| Symbol                                                                                          | Concept                                                                                                                                                                            | First Defined      | Used In  |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------|
| $S1-S9$                                                                                         | Nine numerically registered QT/HT joint-spectral sectors                                                                                                                           | Paper II §2        | II, III  |
| $\mathcal{B}_{QH} = \text{alg}(A, QT_{\text{all}}, HT_{\text{all}})$                            | Conditional exact algebra associated with the numerically commuting QT/HT registration; not identified with the full group commutant                                               | Paper II §2        | II, III  |
| $QT_{\text{all}}, HT_{\text{all}}$                                                              | Quarter-turn / half-turn total averages                                                                                                                                            | Paper II §2        | II, III  |
| $Q_\alpha$                                                                                      | QT/HT joint-spectral sector projector; not assumed $G$ -invariant                                                                                                                  | Paper II §2        | II, III  |
| $K_{\beta\alpha}^S = \max_{s \in S} \ Q_\beta \rho(s) Q_\alpha\ _F$                             | Direct generator-transport norm                                                                                                                                                    | Paper II §3.1      | II, III  |
| $\text{Supp}_{\text{nc}}(\alpha)$                                                               | Thresholded family-level block localizer using the maximum over all three per-axis QT commutator pairs; Type I labels require overlap by definition, but overlap is not sufficient | Paper II §4        | II       |
| $\  [QT^0, QT^1] \ _b$                                                                          | Per-block QT commutator norm                                                                                                                                                       | Paper II §4.2      | II, III  |
| $QT^a$                                                                                          | Per-axis quarter-turn averaging operators                                                                                                                                          | Paper II App A     | II, III  |
| $(a \in \{0, 1, 2\})$                                                                           |                                                                                                                                                                                    |                    |          |
| Type I / Type II transport                                                                      | Post-certification labels for the nine shared-noncommutative-support edges and the one CP exception; not universal sufficient criteria                                             | Paper II §4        | II       |
| CP permutation channel                                                                          | Registered $S8 \leftrightarrow S9$ Type II edge ( $K = 2.83$ )                                                                                                                     | Paper II §4        | II       |
| $A_{EP} \cong M_2(\mathbb{C})^4 \oplus M_1(\mathbb{C})^4$                                       | Computational EP block algebra census (20-dimensional with 8-dimensional center)                                                                                                   | Paper II §4        | II       |
| QH refinement boundary                                                                          | Conditional minimality only inside the declared commuting QH algebra; no global maximal-refinement theorem                                                                         | Paper II §5        | II       |
| Hub pattern                                                                                     | Sparse ten-edge graph with unique degree-five hub S6; the graph is not a star                                                                                                      | Paper II §3.4      | II, III  |
| S1 isolation                                                                                    | Machine-zero off-diagonal direct transport in the canonical audit; exact $G$ -invariance remains unproved                                                                          | Paper II §4        | II       |
| Generator-family comparison                                                                     | Extended computational census retained in CCS; not part of the Paper II theorem spine                                                                                              | CCS Part II        | CCS only |
| $T_{\beta\alpha}(g) = Q_\beta \rho(g) Q_\alpha$                                                 | Generator transport block from source $\alpha$ to target $\beta$                                                                                                                   | Paper II §3.1      | II, III  |
| $\sum_{\beta \neq \alpha} \ T_{\beta\alpha}(g)\ _F^2 = \frac{1}{2} \ [\rho(g), Q_\alpha]\ _F^2$ | Off-diagonal transport–non-invariance identity                                                                                                                                     | Paper II Prop. 3.5 | II, III  |

## External Independent Paper III

The independent Paper III defines its support graph, projected composition operators, image–kernel obstruction, local promotion criteria, and Rubik matrix certificate in its own manuscript. The CCS does not define, number, or certify that theorem spine.

## Prototypes, Controls, and Cross-Cutting

| Symbol                           | Concept                                                                                                        | First Defined              | Used In         |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------|-----------------|
| $S_3$ nat $\oplus$ reg (9-dim)   | Archived first-version sector-invariance control; not matrix-composition evidence                              | excluded source provenance | provenance only |
| $S_3$ reg $\oplus$ reg (12-dim)  | Archived first-version sector-invariance control; not matrix-composition evidence                              | excluded source provenance | provenance only |
| N=2 pocket cube (72-dim)         | Archived first-version graph/kappa control; not matrix-composition evidence                                    | CCS legacy control         | provenance only |
| Archived S1–S6 summaries         | First-version empirical summaries; current status is determined item by item                                   | CCS legacy §II.5           | archive only    |
| Cross-paper comparison path      | Spectral sectors $\rightarrow$ direct support graph $\rightarrow$ projected composition audit                  | Independent Papers I–III   | comparison only |
| Canonical layer keys             | $\lambda = 1 - k/9$ :<br>[1, 8/9, 7/9, 2/3, 5/9, 1/3]                                                          | CCS Part 0.5               | I, II, III      |
| $m =  S /2$                      | Effective generator count                                                                                      | Paper I §7.3               | I               |
| Face-symmetric / symmetry-broken | Generator family classification                                                                                | Paper I §7.4               | I, II           |
| Registered arithmetic contrast   | Historical generator-family scans with values numerically recognized in $\mathbb{Q}$ or $\mathbb{Q}(\sqrt{5})$ | CCS Part II                | archive only    |

## Terminology Convention

| Term                               | Definition                                                                                                                                                                                                                                                                  |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QT/HT joint-spectral sector</b> | One of the nine numerical joint-spectral clusters registered from the declared QT/HT averages. Under exact commuting-Hermitian registration, these become the joint spectral subspaces associated with primitive spectral idempotents of the generated commutative algebra. |
| <b>hybrid sector</b>               | A QT/HT joint-spectral sector whose projector has nonzero support on more than one block. There are 6 hybrid sectors: S1 (cp+ep), S3 (ep+eo), S4 (ep+co), S6 (ep+eo), S7 (cp+ep+co+eo), S9 (cp+co). S7 is the unique all-block hybrid spanning all four blocks.             |
| <b>transport-active</b>            | A sector pair $(\alpha, \beta)$ is transport-active if $K_{\alpha\beta} > 0$ (non-zero one-step transport). All 10 direct edges are block-preserving.                                                                                                                       |
| <b>registered</b>                  | The nine-sector numerical decomposition obtained from the declared QT/HT pair.                                                                                                                                                                                              |
| <b>QH sectorization</b>            | Exact commutation is a hypothesis for the corresponding algebraic joint-resolution statement. It is not a full- $G$ commutant decomposition.                                                                                                                                |

**Geometric & move conventions** (coordinate system, cubie ordering, generator encoding, action direction, block decomposition, numerical tolerances) are maintained in `docs/conventions.md`.

## Part 0.5 — Registered API Surface

**Purpose.** Record the current mapping between mathematical notation, computational interfaces, and archived numerical outputs. This section answers which implementation produced a value; it does not make that implementation a mathematical definition.

**Scope.** Functions listed here are the registered v2 implementation entry points. Alternative implementations may be used when their conventions, parameters, and comparison residuals are declared.

**Dependencies.** `rime.cubieoperator.CubieSpectralOperator` (primary), `rime.cubie.CubieMove` (generator enumeration), `rime.spectral_utils` ( $S_3$  negative controls, joint diagonalization helpers).

**Outputs.** All numerical values in CCS Parts I–II are produced by the functions listed below.

*This part records the current paper-to-code mapping; executable scripts and structured artifacts remain the computational certificate layer.*

### 0.5.1 Spectral Objects

| Mathematical object      | Canonical API                                   | Returns                                                                                        | Stability |
|--------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------|-----------|
| Layer eigenvalues        | <code>CubieSpectralOperator().layer_keys</code> | <code>list[float]</code> — 6 canonical $\lambda$ , descending (property, not method)           | <b>A</b>  |
| Layer dimension          | <code>.layer_dimension(lam)</code>              | <code>int</code>                                                                               | <b>A</b>  |
| Layer projector          | <code>.layer_projector(lam)</code>              | <code>ndarray</code> (228×228)                                                                 | <b>A</b>  |
| Closest layer            | <code>.closest_layer(lam)</code>                | <code>float</code> — canonical $\lambda$ key                                                   | <b>A</b>  |
| Sector decomposition     | <code>.center_decomposition()</code>            | <code>dict</code> with <code>n_sectors</code> , <code>projectors</code> , <code>sectors</code> | <b>A</b>  |
| A <sub>18</sub> operator | <code>.A</code> (property)                      | <code>ndarray</code> (228×228)                                                                 | <b>A</b>  |

### 0.5.2 Transport and Accessibility

| Mathematical object                       | Canonical API                                                  | Returns                                                                                     | Stability  |
|-------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------|
| Direct-support matrix plus legacy arrays  | <code>.transport_kappa(projectors, compute_kappa1=True)</code> | <code>tuple[K, kappa0, kappa1]</code> ; only $K$ is the current direct-support object       | <b>B/C</b> |
| First-version $\kappa$ array at depth $d$ | <code>.kappa_depth(d)</code>                                   | archived principal-log diagnostic; not current exact Lie depth                              | <b>C</b>   |
| Principal-log registration                | <code>.compute_lie_generators()</code>                         | <code>list[ndarray]</code> — 18 numerically skew-Hermitian matrices for the declared branch | <b>B/C</b> |
| $\rho(g)$ matrices                        | <code>.rho_matrices()</code>                                   | <code>list[ndarray]</code> — 18 unitary representation matrices                             | <b>A</b>   |



### 0.5.3 Algebraic Structure

| Mathematical object                  | Canonical API                                           | Returns                                                                                    | Stability |
|--------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------|
| Ambient commutant candidate          | <code>.full_commutant_combinatorial()</code>            | numerical/combinatorial candidate basis of dimension 610; exact certificate still required | <b>C</b>  |
| Compressed spectral-layer commutants | <code>.commutant_algebra()</code>                       | withdrawn layerwise interpretation; archive compatibility only                             | <b>C</b>  |
| Block projectors                     | <code>BLOCK_RANGES</code> (in <code>rime.cubie</code> ) | block index slices                                                                         | <b>A</b>  |
| QT/HT per-axis ops                   | <code>.build_per_axis_ops()</code>                      | $QT^0, QT^1, QT^2, HT^0, HT^1, HT^2$                                                       | <b>A</b>  |

### 0.5.4 $S_3$ Prototypes

| Mathematical object                | Canonical API                                                     | Returns                                                           | Stability |
|------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|-----------|
| $S_3$ representations              | <code>rime.spectral_utils.build_s3*_rep()</code>                  | <code>ndarray</code> — $S_3$ irrep matrices                       | <b>A</b>  |
| Joint diagonalization              | <code>rime.spectral_utils.joint_diag_sectors()</code>             | sector projectors                                                 | <b>A</b>  |
| Two-step projected product maximum | <code>rime.spectral_utils.max_two_step_composition()</code>       | maximum Frobenius norm and maximizing generator pair              | <b>B</b>  |
| Graph-only candidate enumeration   | <code>rime.spectral_utils.find_graph_only_two_step_pairs()</code> | endpoint/intermediate triples requiring matrix audit              | <b>B</b>  |
| First-version T7 detector          | <code>rime.spectral_utils.find_t7_pairs()</code>                  | archived support-level interpretation; not a morphism certificate | <b>C</b>  |

### 0.5.5 Generator Enumeration

| Mathematical object     | Canonical API                                   | Returns                      | Stability |
|-------------------------|-------------------------------------------------|------------------------------|-----------|
| 18 face-turn generators | <code>CubieMove.prim_moves</code>               | <code>list[CubieMove]</code> | <b>A</b>  |
| Generator weighting     | <code>A_18 = (12 QT_all + 6 HT_all) / 18</code> | Definitional identity        | <b>A</b>  |

**Legacy stability key:** **A** = invariant under the listed recomputation, permutation, and gauge checks; **B** = stable with fixed parameters and the declared tolerance sweep; **C** = exploratory or withdrawn. These tags describe the archived implementation record and are not the current four-level paper claim status.

**Implementation boundary.** The listed APIs are the registered implementation paths used to generate this archive. Alternative implementations are allowed when they declare conventions, parameters, tolerances, and comparison residuals. The independent papers and their claim-specific executable artifacts, not this API table, control current claims and certificates.

## Part I — Core Numerical Structures

**Purpose.** Register the numerical objects used by Papers I–II. These data do not replace paper-level proofs and do not govern revised Paper III.

**Scope.** Operators, eigenspaces, sectors, projectors at 6-layer ( $A_{18}$ ) and 9-sector (Center) resolution. All canonical tables live in this Part.

**Dependencies.** The Rubik’s cube representation construction (`rime/cubie.py`, `rime/cubieoperator.py`).

**Outputs.** All objects and numerical values referenced by Parts II–III and the papers.

**CCS Fig. C0 omitted.** The first-version combined pipeline is not part of the current reproducibility compendium.

### 1.1 Representation Space

The Rubik’s cube group acts on a 228-dimensional complex vector space with four G-invariant blocks:

$$V = V_{\text{cp}} \oplus V_{\text{ep}} \oplus V_{\text{co}} \oplus V_{\text{eo}}, \quad 64 + 144 + 8 + 12 = 228$$

**Table C1 — Block Decomposition.**

| Block | Dim | Content                               | Algebra                                                                                    |
|-------|-----|---------------------------------------|--------------------------------------------------------------------------------------------|
| CP    | 64  | Corner permutation                    | $Q_3$ Hamming scheme $H(3,2)$ ,<br>Bose-Mesner algebra $\cong$ Hecke $H(S_2 \wr S_3, S_3)$ |
| EP    | 144 | Edge permutation                      | Face-incidence adjacency $JJ^T$ ,<br>noncommutative core                                   |
| CO    | 8   | Corner orientation ( $\mathbb{Z}_3$ ) | Abelian phase structure                                                                    |
| EO    | 12  | Edge orientation ( $\mathbb{Z}_2$ )   | Abelian phase structure                                                                    |

Block order throughout:  $\text{CP} \rightarrow \text{EP} \rightarrow \text{CO} \rightarrow \text{EO}$ .

### 1.2 Averaging Operator

$$A = \frac{1}{|S|} \sum_{s \in S} \rho(s)$$

For the canonical 18 face-turn generators ( $S = S^{-1}$ ):

$$A_{18} = (12 \text{QT}_{\text{all}} + 6 \text{HT}_{\text{all}})/18$$

where  $\text{QT}_{\text{all}} = \sum_{a \in \{x,y,z\}} \text{QT}^a$ ,  $\text{HT}_{\text{all}} = \sum_{a \in \{x,y,z\}} \text{HT}^a$ , and

$$\text{QT}^a = \frac{1}{2}(\rho(+a) + \rho(-a)), \quad \text{HT}^a = \rho(2a)$$

$A$  is Hermitian because the declared generator family is inverse closed and the representation is unitary; Paper I states the general result with its hypotheses.

In the declared complex128 realization, the QT/HT averages are registered as numerically commuting. Conditional on exact commutation, the corresponding commutative algebra is

$$Z_{\text{QH}} = \langle A_{18}, \text{QT}_{\text{all}}, \text{HT}_{\text{all}} \rangle = \langle \text{QT}_{\text{all}}, \text{HT}_{\text{all}} \rangle.$$

The nine sectors in §1.4 are numerical joint-spectral clusters. Conditional on exact registration, the six displayed layers are the collision quotient obtained by the linear projection

$$L_{2/3}(q, h) = (2q + h)/3.$$

### 1.3 Six Canonical Layers

The declared computation registers six eigenspaces of  $A_{18}$  against the displayed rational values  $\lambda = 1 - k/9$ . Conditional on exact QT/HT registration, these are the collision quotients of the nine joint-spectral sectors under  $L_{2/3}$ .

**Table C2 — Six Canonical Layers.**

| $k$ | $\lambda$ | dim | Label     | Block composition               | Layer |
|-----|-----------|-----|-----------|---------------------------------|-------|
| 0   | 1         | 20  | $V_1$     | cp(8) + ep(12)                  | A     |
| 1   | 8/9       | 2   | $V_{8/9}$ | eo(2)                           | A     |
| 2   | 7/9       | 39  | $V_{7/9}$ | ep(36) + eo(3)                  | A     |
| 3   | 2/3       | 26  | $V_{2/3}$ | ep(24) + co(2)                  | A     |
| 4   | 5/9       | 106 | $V_{5/9}$ | cp(24) + ep(72) + co(3) + eo(7) | A     |
| 6   | 1/3       | 35  | $V_{1/3}$ | cp(32) + co(3)                  | A     |

$k = 5$  ( $\lambda = 4/9$ ) is absent from the registered block census; see §1.7 for the exact cp/ep reductions and qualified co/eo audit.

Canonical layer keys:  $[1, 8/9, 7/9, 2/3, 5/9, 1/3]$  ( $\lambda = 1 - k/9$ ,  $k \in \{0, 1, 2, 3, 4, 6\}$ ).

### 1.4 Nine QT/HT Joint-Spectral Sectors

These nine numerical clusters are produced by the declared QT/HT joint diagonalization and registration policy. For commuting Hermitian QT/HT operators, the corresponding projectors are primitive spectral idempotents of the generated commutative algebra. This does not assert a finest orthogonal sectorization in all of  $\text{End}(V)$ .

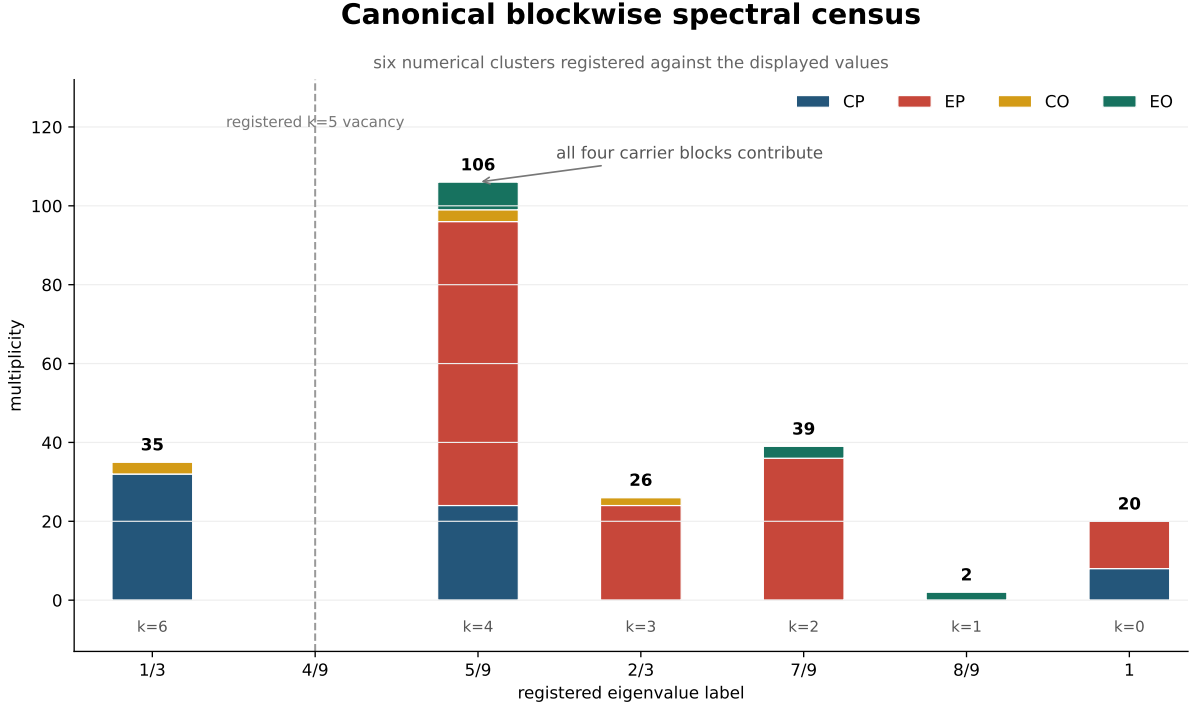
**Table C3 — Nine QT/HT Joint-Spectral Sectors.**

| Sector | dim | $k$ | $\lambda_{18}$ | $\lambda_{\text{QT}}$ | $\lambda_{\text{HT}}$ | Block support             | Layer     | Role               |
|--------|-----|-----|----------------|-----------------------|-----------------------|---------------------------|-----------|--------------------|
| S1     | 20  | 0   | 1              | 1                     | 1                     | cp(8)+ep(12)              | $V_1$     | ISOLATED           |
| S2     | 2   | 1   | 8/9            | 5/6                   | 1                     | eo(2)                     | $V_{8/9}$ | Connective         |
| S3     | 39  | 2   | 7/9            | 5/6                   | 2/3                   | ep(36)+eo(3)              | $V_{7/9}$ | Metastable         |
| S4     | 26  | 3   | 2/3            | 1/2                   | 1                     | ep(24)+co(2)              | $V_{2/3}$ | Intermediate       |
| S5     | 1   | 4   | 5/9            | 1/3                   | 1                     | eo(1)                     | $V_{5/9}$ | Tiny EO            |
| S6     | 39  | 4   | 5/9            | 1/2                   | 2/3                   | ep(36)+eo(3)              | $V_{5/9}$ | <b>PRIMARY HUB</b> |
| S7     | 66  | 4   | 5/9            | 2/3                   | 1/3                   | cp(24)+ep(36)+co(3)+eo(3) | $V_{5/9}$ | Secondary hub      |
| S8     | 8   | 6   | 1/3            | 0                     | 1                     | cp(8)                     | $V_{1/3}$ | Pure CP            |
| S9     | 27  | 6   | 1/3            | 1/3                   | 1/3                   | cp(24)+co(3)              | $V_{1/3}$ | CP+CO              |

Sector ordering: CCS canonical — sort by  $k = 9(1 - \lambda_{18})$  ascending, then by dimension ascending within fixed  $k$ . Labels S1–S9 frozen by this table. Raw joint diagonalization yields 11 sectors;

S4+S5 and S9+S10 are merged based on coincident eigenvalue triples (gap  $> 10^{-3}$  between genuinely distinct triples).

$V_{5/9}$  splits into 3 sectors (S5, S6, S7).  $V_{1/3}$  splits into 2 sectors (S8, S9). Thus the six-layer  $A_{18}$  decomposition is a coarse collision quotient of the nine-sector QT/HT joint spectrum.



**CCS Figure C1.** Current source-addressed blockwise spectral census. The six numerical clusters are registered against the displayed values; per-block status remains as stated in §1.5.

## 1.5 Block-Level Spectral Derivations

This section preserves the finite block reductions and orientation-block audits behind the registered census. The cp and ep reductions are exact combinatorial calculations. The co and eo sections retain their explicit numerical inputs. Paper I, rather than this archive, owns the block-union theorem and its proof.

### 1.5.1 The cp Block: $Q_3$ Hypercube Bose–Mesner Algebra

The 8 corner positions are the vertices of a 3-dimensional hypercube  $Q_3$  with coordinates  $\{\pm 1\}^3$ . Each face turn cycles the 4 corners on that face. The corner-permutation representation factors as:

$$\rho_{cp}(s) = P_{\text{perm},8}(s) \otimes I_8$$

where  $P_{\text{perm},8}(s)$  is the  $8 \times 8$  permutation matrix of the corner positions, and  $I_8$  acts on the internal orientation label at each position.

Define the position transition sum  $S_8 = \sum_{s \in S} P_{\text{perm},8}(s)$ . For the 18-full family, the entry  $S_8[i, j]$  depends only on the Hamming distance in  $Q_3$ :

$$S_8 = 9I + 2A_1 + A_2$$

where  $A_k$  is the distance- $k$  adjacency of  $Q_3$ . The coefficients reflect the geometry:

- A corner is fixed by the 3 faces not incident to it:  $3 \times 3 = 9$  (diagonal)
- Adjacent corners (Hamming distance 1) share 2 faces, each contributing a quarter-turn sending one to the other: 2
- Face-diagonal corners (Hamming distance 2) share 1 face, with the  $180^\circ$  turn providing the transition: 1
- Cube-diagonal corners (Hamming distance 3) share no face: 0

The eigenfunctions of  $Q_3$  are indexed by binary vectors  $u \in \{0, 1\}^3$  with  $v_u[x] = (-1)^{u \cdot x}$ . The eigenvalue of  $A_k$  on  $v_u$  depends only on  $|u|$  (Hamming weight):

$$\begin{aligned} |u| = 0 : A_1 v_u &= 3v_u, A_2 v_u = 3v_u &\Rightarrow S_8 v_u &= (9 + 6 + 3)v_u = 18v_u &(\times 1) \\ |u| = 1 : A_1 v_u &= 1v_u, A_2 v_u = -1v_u &\Rightarrow S_8 v_u &= (9 + 2 - 1)v_u = 10v_u &(\times 3) \\ |u| = 2 : A_1 v_u &= -1v_u, A_2 v_u = -1v_u &\Rightarrow S_8 v_u &= (9 - 2 - 1)v_u = 6v_u &(\times 3) \\ |u| = 3 : A_1 v_u &= -3v_u, A_2 v_u = 3v_u &\Rightarrow S_8 v_u &= (9 - 6 + 3)v_u = 6v_u &(\times 1) \end{aligned}$$

Hence  $\text{Spec}(S_8) = \{18^{(1)}, 10^{(3)}, 6^{(4)}\}$ . With  $A_{\text{cp}} = (1/18)S_8 \otimes I_8$ , the eigenvalues are  $(1/18) \times \{18, 10, 6\} = \{1, 5/9, 1/3\}$ . Using  $k = 9(1 - \lambda)$ :

$$\mathcal{K}_{\text{cp}} = \{0, 4, 6\}, \quad \text{multiplicities: } 8 \times (1, 3, 4) = (8, 24, 32)$$

The Bose–Mesner algebra of  $Q_3$  is the Hamming association scheme  $H(3, 2)$ , isomorphic to the Hecke algebra  $H(S_2 \wr S_3, S_3)$ . The Krawtchouk polynomials  $K_k(i; 3, 2)$  give the eigenvalues of  $A_i$  on the  $k$ -th eigenspace, providing the closed-form spectral character table.

### 1.5.2 The ep Block: Face-Incidence Adjacency Algebra

The 12 edge positions and 6 faces define a  $12 \times 6$  edge-face incidence matrix  $J$ :  $J[e, F] = 1$  if edge  $e$  lies on face  $F$ . Each edge belongs to exactly 2 faces; each face contains exactly 4 edges.

The edge-permutation representation factors as:

$$\rho_{\text{ep}}(s) = P_{\text{perm}, 12}(s) \otimes I_{12}$$

For the 18-full family, every move on face  $F$  cycles its 4 edges. Two edges share a face iff they are both on at least one common face, giving:

$$S_{12} = 10I + JJ^\top$$

(The term  $10I$ : each edge lies on 2 faces; the 4-cycle on each face contributes 2 moves sending the edge to a different position plus 1 move (the  $180^\circ$ ) possibly sending it back; detailed counting yields 10 fixed-point contributions.)

The nonzero eigenvalues of  $JJ^\top$  are those of the  $6 \times 6$  Gram matrix  $J^\top J = 4I + A_{\text{face}}$ , where  $A_{\text{face}}$  is the adjacency matrix of the cube's face graph — the octahedron graph on 6 vertices, where two faces are adjacent if they share an edge.

The opposite-face permutation  $P$  pairs each face with its antipode; then  $A_{\text{face}} = J_6 - I - P$ . Since  $P^2 = I$ , the common eigenvectors of  $J_6$  and  $P$  give:

$$\text{Spec}(A_{\text{face}}) = \{4^{(1)}, 0^{(3)}, -2^{(2)}\}, \quad \text{Spec}(J^\top J) = \{8^{(1)}, 4^{(3)}, 2^{(2)}\}$$

Projecting back to the 12-dimensional edge space adds a 6-dimensional nullspace:

$$\text{Spec}(JJ^\top) = \{8^{(1)}, 4^{(3)}, 2^{(2)}, 0^{(6)}\}, \quad \text{Spec}(S_{12}) = \{18^{(1)}, 14^{(3)}, 12^{(2)}, 10^{(6)}\}$$

With  $(1/18)S_{12}$  eigenvalues  $\{1, 7/9, 2/3, 5/9\}$ , we obtain:

$$\mathcal{K}_{\text{ep}} = \{0, 2, 3, 4\}, \quad \text{multiplicities: } 12 \times (1, 3, 2, 6) = (12, 36, 24, 72)$$

The matrix  $JJ^\top$  generates a finite commutative adjacency algebra. This archive does not formally identify that algebra with a specific new association scheme; reconstructing the relevant coherent-configuration or association-scheme object remains an open combinatorial problem.

### 1.5.3 The co Block: Symmetry-Guided Computation

The corner-orientation block is the only block where generator matrix entries live in  $\mathbb{Z}[\omega]$  rather than  $\mathbb{Z}$ , with  $\omega = e^{2\pi i/3}$ . Cube symmetry constrains the possible multiplicities, but it does not force the displayed accidental degeneracy. The final spectrum therefore remains a symmetry-guided computational proposition.

**Computational Proposition (registered CO spectrum).** Let  $A_{\text{co}} = \frac{1}{18} \sum_{s \in S} \rho_{\text{co}}(s)$  for the declared 18 face-turn realization. The symmetry decomposition and direct matrix audit register:

1. The permutation representation of the cube symmetry group  $O$  on the 8 corners decomposes as  $\chi_{\text{corners}} = A_1 \oplus A_2 \oplus T_1 \oplus T_2$  (irrep dimensions  $1 + 1 + 3 + 3 = 8$ ).
2. By Schur's lemma,  $A_{\text{co}}$  acts as a scalar on each irreducible  $O$ -submodule:

$$A_{\text{co}} = \lambda_{A_1} P_{A_1} + \lambda_{A_2} P_{A_2} + \lambda_{T_1} P_{T_1} + \lambda_{T_2} P_{T_2}$$

3. The spectrum is:

$$\text{Spec}(A_{\text{co}}) = \left\{ \frac{2}{3}, \frac{2}{3}, \frac{5}{9}^{(3)}, \frac{1}{3}^{(3)} \right\}, \quad \mathcal{K}_{\text{co}} = \{3, 4, 6\}, \quad (d_3, d_4, d_6) = (2, 3, 3)$$

#### Audit sketch.

*Diagonal & trace.*  $\text{Tr}(\rho_{\text{co}}(g)) = 4$  for all 18 generators: each face turn fixes the 4 corners on the opposite face (no orientation change  $\rightarrow +1$  contribution for each fixed corner). Hence  $\text{Tr}(A_{\text{co}}) = 4$  and  $A_{\text{co}}[i, i] = 9/18 = 1/2$  (each corner is fixed by 9 generators: 6 on the two faces containing it, plus 3 opposite-face half-turns that preserve orientation).

*$O_h$  invariance.* The set of 18 face-turn generators is closed under cube symmetries. Therefore  $A_{\text{co}}$  is  $O_h$ -invariant and respects the  $O$ -irrep decomposition of the 8-dimensional corner permutation representation.

*Adjacency structure.* Work with  $M_{\text{co}} = 18(A_{\text{co}} - I/2)$ , whose off-diagonal entries are determined purely by cube geometry. Three adjacency classes emerge:

| Class         | Shared faces | Pairs per corner | $M_{\text{co}}[i, j]$          | Count |
|---------------|--------------|------------------|--------------------------------|-------|
| Edge-adjacent | 2            | 2                | $1 + \omega$ or $1 + \omega^2$ | 8     |
| Face-opposite | 1            | 4                | $\pm 1$                        | 16    |
| Body-opposite | 0            | 1                | 0                              | 4     |

Total:  $8 + 16 + 4 = 28 = \binom{8}{2}$ . Each corner has  $2 + 4 + 1 = 7$  neighbours. ✓

*Row sum  $\rightarrow A_1$  eigenvalue.* The row sum of  $M_{\text{co}}$  is uniform ( $= 3$ ). The imaginary parts of the edge-adjacent entries cancel ( $\omega + \omega^2 = -1$ ), and the  $\pm 1$  real entries sum to give net  $+3$  after accounting for the adjacency structure. Hence:

$$\lambda_{A_1} = \frac{1}{2} + \frac{3}{18} = \frac{2}{3} \quad (k = 3)$$

$M_{\text{co}}$  spectrum. Diagonalizing the Hermitian,  $O_h$ -invariant matrix  $M_{\text{co}}$ :

$$\text{Spec}(M_{\text{co}}) = \{3^{(2)}, 1^{(3)}, -3^{(3)}\}$$

Converting to  $A_{\text{co}}$  eigenvalues:

$$\lambda = \frac{1}{2} + \frac{\mu}{18} : \quad \mu = 3 \mapsto \lambda = \frac{2}{3} \quad (k = 3), \quad \mu = 1 \mapsto \lambda = \frac{5}{9} \quad (k = 4), \quad \mu = -3 \mapsto \lambda = \frac{1}{3} \quad (k = 6)$$

*Accidental  $A_1/A_2$  degeneracy.* The multiplicity-2 eigenvalue at  $\mu = 3$  implies  $\lambda_{A_1} = \lambda_{A_2} = 2/3$  — both 1-dimensional  $O$ -irreps carry the same eigenvalue. This is not forced by any obvious symmetry (the two 1-dim irreps could in principle carry distinct eigenvalues) but is verified numerically to machine precision.

*Irrep assignment.* The eigenvalue-multiplicity pattern  $(2, 3, 3)$  matches the  $O$ -irrep dimensions  $(1, 1, 3, 3)$  with the accidental degeneracy  $1 + 1 = 2$ :

| $\lambda$ | $k$ | mult | $O$ -irrep           |
|-----------|-----|------|----------------------|
| $2/3$     | 3   | 2    | $A_1 \oplus A_2$     |
| $5/9$     | 4   | 3    | $T_1$ or $T_2$       |
| $1/3$     | 6   | 3    | the other $T$ -irrep |

(The isotypic assignment of the two 3-dimensional  $T$ -irreps to  $5/9$  vs  $1/3$  is not resolved.)

*Trace consistency.*  $2 \cdot \frac{2}{3} + 3 \cdot \frac{5}{9} + 3 \cdot \frac{1}{3} = \frac{4}{3} + \frac{5}{3} + 1 = 4 = \text{Tr}(A_{\text{co}})$ . ✓

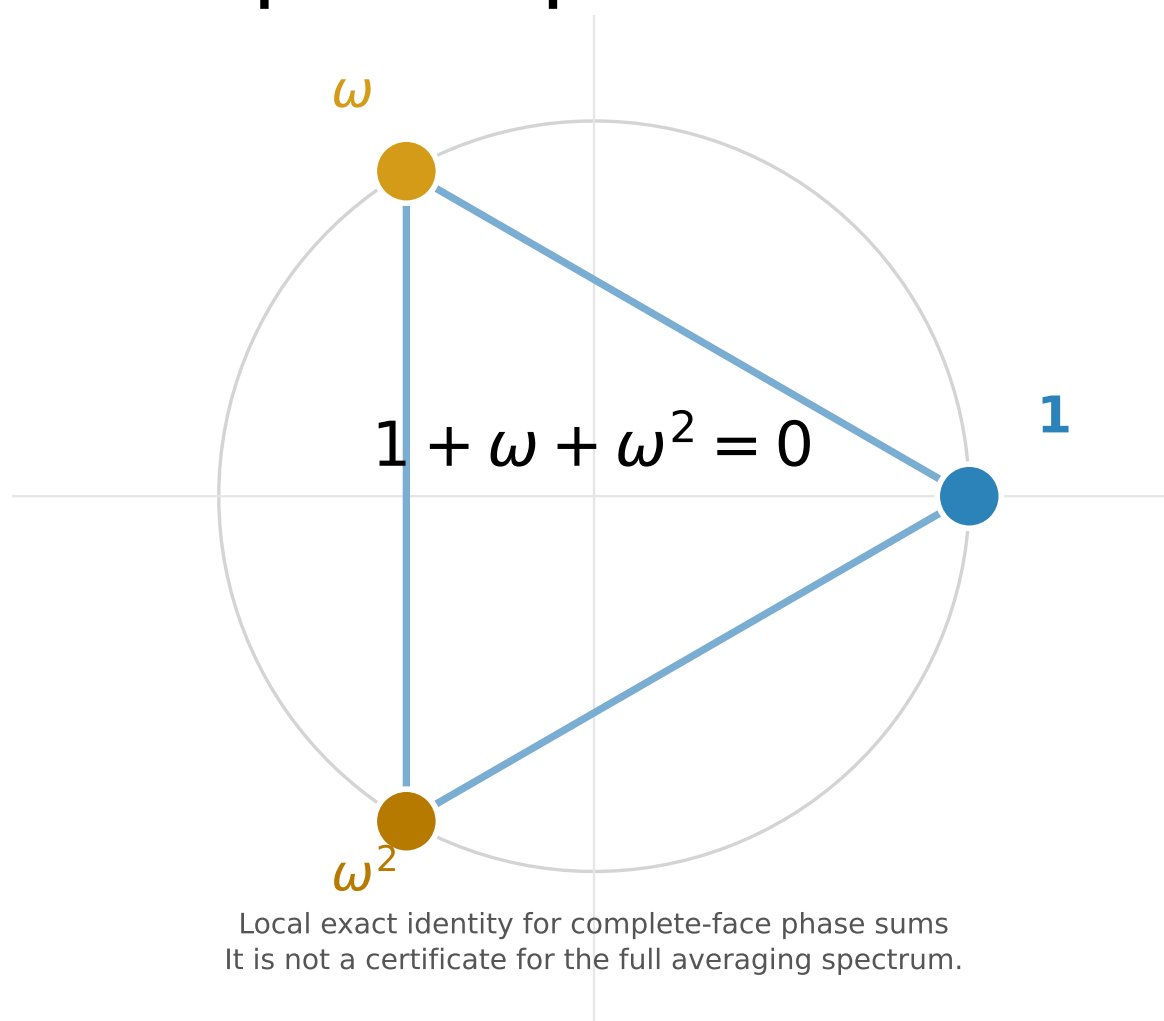
*Local arithmetic identity.* The complete-face phase sum satisfies  $\omega + \omega^2 + 1 = 0$ . This exact local cancellation is compatible with the displayed rational labels, but it is not by itself a compression-trace certificate or a proof of rationality for the full averaging spectrum.

**Status.** Computational proposition. The symmetry decomposition and trace identities are exact local ingredients; the accidental  $A_1/A_2$  degeneracy and the assignment of the two three-dimensional components remain numerical inputs.

#### 1.5.4 The eo Block: Numerical-Representation Observation

The edge-orientation block carries a  $\mathbb{Z}_2$  permutation@phase structure: generators act as monomial matrices with entries in  $\{0, \pm 1\}$  that permute edge positions and flip orientation signs.

## Complete-face phase cancellation



**CCS Figure C10.** Current rendering of the exact local roots-of-unity identity, with its full-spectrum boundary stated in the figure.



The present archive does not supply a complete group-theoretic derivation from  $O_h$  symmetry and Schur's lemma because the isotypic decomposition contains a multiplicity-2 component. The registered spectrum below is therefore a **numerical-representation observation**: it is empirically rigid and structurally consistent, but not an exact theorem.

**Observed spectrum (18-full).**

$$\text{Spec}(A_{\text{eo}}) = \left\{ \frac{8}{9}^{(2)}, \frac{7}{9}^{(3)}, \frac{5}{9}^{(7)} \right\}, \quad \mathcal{K}_{\text{eo}} = \{1, 2, 4\}, \quad (d_1, d_2, d_4) = (2, 3, 7)$$

**Diagonal & trace.**  $\text{Tr}(\rho_{\text{eo}}(g)) = 8$  for all 18 generators (each face turn fixes 8 edges: 4 on the opposite face + 4 equatorial). Hence  $\text{Tr}(A_{\text{eo}}) = 8$  and  $A_{\text{eo}}[i, i] = 12/18 = 2/3$  (each edge is on 2 faces  $\rightarrow$  12 generators fix it, 6 move it). Trace consistency:  $2 \cdot \frac{8}{9} + 3 \cdot \frac{7}{9} + 7 \cdot \frac{5}{9} = 8$ . ✓

**Off-diagonal structure.** All off-diagonal entries are purely real ( $\pm 1/18$ ). The  $Z_2$  orientation phase combined with the permutation action produces only real coupling after averaging. Define  $N_{\text{eo}} = 18(A_{\text{eo}} - 2I/3)$  with entries in  $\{-1, 0, +1\}$ . Each edge couples to exactly 6 others and has zero coupling to 5.

**Two edge classes.** Two distinct edge types emerge from the row sums of  $A_{\text{eo}}$ :

| Class  | Count   | Row sum                                    | Positive couplings | Negative couplings |
|--------|---------|--------------------------------------------|--------------------|--------------------|
| Type A | 4 edges | $1 = \frac{2}{3} + \frac{6}{18}$           | 6                  | 0                  |
| Type B | 8 edges | $\frac{7}{9} = \frac{2}{3} + \frac{2}{18}$ | 4                  | 2                  |

The 4 Type A edges correspond to the 4 space diagonals of the cube; the 8 Type B edges are the remaining edges. This two-class split is  $O_h$ -equivariant — edge classes form orbits under the geometric cube symmetry.

**$N_{\text{eo}}$  spectrum.**

$$\text{Spec}(N_{\text{eo}}) = \{4^{(2)}, 2^{(3)}, -2^{(7)}\}$$

Converting:  $\lambda = \frac{2}{3} + \frac{\mu}{18}$  gives the  $A_{\text{eo}}$  spectrum above.

**Why this is NOT a theorem.** The obstruction is the  $2T_2$  **multiplicity fiber**. Under  $O$ , the permutation representation on 12 edges is conjectured to decompose as  $A_1 \oplus E \oplus T_1 \oplus 2T_2$ . The component  $2T_2$  has multiplicity 2 — by Schur's lemma, an  $O_h$ -invariant operator on an isotypic component of multiplicity  $m > 1$  acts as  $I_m \otimes B$  where  $B$  is a  $(\dim_{\text{irrep}} \times \dim_{\text{irrep}})$  matrix, NOT necessarily scalar. Without explicitly block-diagonalizing the multiplicity fiber, a single eigenvalue cannot be assigned to the  $2T_2$  component using representation theory alone.

A complete analytic proof would require: edge incidence algebra on the signed line graph of the cube, Hecke-type structure encoding the  $Z_2$  orientation representation, and multiplicity-algebra machinery to resolve the  $2T_2$  fiber. These extend beyond the current mathematical framework.

**Generator-family scope.** The registered k-set  $\{1, 2, 4\}$  is specific to the declared 18-full family. Other archived generator families produce different numerical spectra. No exact family-wide rigidity theorem is claimed.

### 1.5.5 Block Spectra Across All Generator Families

**Historical computational observation.** This table preserves a first-version family scan. Its rational and quadratic labels are numerical recognitions, not exact spectral-field certificates. Every family must be recomputed and supplied with an exact characteristic or minimal polynomial before its field label can be promoted.

**Table C4 — Block Spectra Across Generator Families.**

| Family        | $m$  | $\mathcal{K}_{\text{cp}}$ | $\mathcal{K}_{\text{ep}}$ | $\mathcal{K}_{\text{co}}$ | $\mathcal{K}_{\text{eo}}$ | $\mathcal{K}(A)$                  | #layers        |
|---------------|------|---------------------------|---------------------------|---------------------------|---------------------------|-----------------------------------|----------------|
| 18-full       | 9    | $\{0, 4, 6\}$             | $\{0, 2, 3, 4\}$          | $\{3, 4, 6\}$             | $\{1, 2, 4\}$             | $\{0, 1, 2, 3, 4, 6\}$            | 6              |
| 12-quarter    | 6    | $\{0, 2, 4, 6\}$          | $\{0, 1, 2, 3\}$          | $\{2, 3, 4\}$             | $\{0, 2, 4\}$             | $\{0, 1, 2, 3, 4, 6\}$            | 6              |
| 6-half        | 3    | $\{0, 2\}$                | $\{0, 1, 2\}$             | $\{0\}$                   | $\{0\}$                   | $\{0, 1, 2\}$                     | 3              |
| 10-partial    | 5    | $\{0, 2, 3, 4\}$          | $\{0, 1, 2, 3\}$          | $\{0\}$                   | $\{0\}$                   | $\{0, 1, 2, 3, 4\}$               | 5              |
| 21-full+slice | 10.5 | $\{0, 2, 6\}$             | $\{0, 3, 4, 5\}$          | $\{1, 3, 5\}$             | $\{1, 3, 5\}$             | $\{0, 1, 2, 3, 4, 5, 6\}^\dagger$ | 6 <sup>†</sup> |

<sup>†</sup> For 21-full+slice,  $m = 10.5$  (half-integer). The eigenvalue formula becomes  $\lambda = 1 - 2k/21$ . The union  $\mathcal{K}(A) = \{0, 1, 2, 3, 4, 5, 6\}$  contains 7 candidate values;  $k = 1$  is eliminated by trace integrality constraint C4 (§7.2), leaving 6 observed eigenvalues. In the  $|S|$ -denominator convention, the k-set is  $\{0, 4, 6, 8, 10, 12\}$  (all even).

#### Key structural observations:

1. **Block profiles determine k.** Each admissible k-value corresponds to a specific combination of active blocks. The block profile is a sharper invariant than the k-value itself: the same k can appear in different families with different block profiles (e.g.,  $k = 2$  in 18-full is ep+eo, while  $k = 2$  in 12-quarter is cp+ep+eo).
2. **The co block is the decisive arithmetic filter.** Within this archived family census, it is the only block whose generator matrix entries lie in  $\mathbb{Z}[\omega]$  rather than  $\mathbb{Z}$ . An eigenspace can have  $d_{\text{co}} > 0$  only for specific k-values where the  $\omega$ -phase cancellation across complete faces yields integer per-face trace sums.
3. **The number of layers is  $|\mathcal{K}(A)|$ , not  $m + 1$ .** The 6 layers in the 18-full case is not a fundamental constant — it is the size of the admissible k-set for this specific generator family.
4. **Forbidden k-values** are those for which no block-dimension assignment satisfies all integrality constraints (see §7.2 for the full Diophantine system C1–C5).

## 1.6 Blockwise Union and the Six-Layer Census

The declared block computations register repeated eigenvalues across the four physical blocks. Grouping equal displayed values gives the following six global layers. This is a blockwise census, not a claim that a fixed number of primitive idempotents is canonically present across all four block algebras.

**Full resonance merging table (18-full,  $m = 9$ ):**

**Table C5 — Registered Blockwise Union.**

| Global $\lambda$ | $k$ | dim | cp $k$ ( $d$ ) | ep $k$ ( $d$ ) | co $k$ ( $d$ ) | eo $k$ ( $d$ ) | Blocks merged        |
|------------------|-----|-----|----------------|----------------|----------------|----------------|----------------------|
| 1                | 0   | 20  | 0 (8)          | 0 (12)         | —              | —              | cp + ep              |
| 8/9              | 1   | 2   | —              | —              | —              | 1 (2)          | eo only              |
| 7/9              | 2   | 39  | —              | 2 (36)         | —              | 2 (3)          | ep + eo              |
| 2/3              | 3   | 26  | —              | 3 (24)         | 3 (2)          | —              | ep + co              |
| 5/9              | 4   | 106 | 4 (24)         | 4 (72)         | 4 (3)          | 4 (7)          | cp + ep +<br>co + eo |
| 1/3              | 6   | 35  | 6 (32)         | —              | 6 (3)          | —              | cp + co              |

$k = 5$  does not occur in the registered block spectra. Section 1.7 records the status of that absence block by block.

For any block-diagonal operator, the spectrum is the union of the block spectra. In the declared computation, the six displayed labels are the union of the four registered block spectra.

The historical “10 to 6” rendering is retained in the figure archive but is not used in this release because its primitive-idempotent count was not a stable typed object.

### 1.7 The $k = 5$ Vacancy: Blockwise Status

The vacancy at  $k = 5$  is registered across all four blocks. The cp and ep exclusions follow from the exact combinatorial reductions above, whereas the co and eo exclusions remain tied to the displayed computational spectra.

**cp block:** The  $Q_3$  hypercube Bose–Mesner algebra has eigenspaces indexed by Hamming weight  $|u| \in \{0, 1, 2, 3\}$ . The eigenvalue of  $S_8$  on  $|u|$  is  $\lambda_{|u|} = 1 - k_{|u|}/9$  where:

- $|u| = 0$ :  $S_8 = 18 \Rightarrow k = 0$
- $|u| = 1$ :  $S_8 = 10 \Rightarrow k = 4$
- $|u| = 2, 3$ :  $S_8 = 6 \Rightarrow k = 6$

The Krawtchouk polynomial  $K_k(i; 3, 2)$  has no root configuration that would produce  $k = 5$ . The cp block spectrum is  $\{0, 4, 6\}$  —  $k = 5$  is Krawtchouk-incompatible.

**ep block:** The face-incidence adjacency algebra has eigenvalues of  $S_{12}$ :

$$\text{Spec}(S_{12}) = \{18, 14^{(3)}, 12^{(2)}, 10^{(6)}\}$$

Converting to k-values:  $k = 9(1 - \lambda/18)$  gives  $\{0, 2, 3, 4\}$ . The octahedron graph spectrum  $\{4, 0^{(3)}, -2^{(2)}\}$  forces exactly these four k-values. No linear combination of the scheme’s adjacency matrices yields  $k = 5$ .

**co block:** The  $\mathbb{Z}_3$  permutation@phase structure yields  $\mathcal{K}_{\text{co}} = \{3, 4, 6\}$ . The phase cancellation  $\omega + \omega^2 + 1 = 0$  restricts the co spectrum to k-values where the accumulated  $\mathbb{Z}_3$  character sums are integer-valued.  $k = 5$  would require a fractional  $\omega$ -phase contribution that cannot be cancelled by any face-complete generator combination.

**eo block:** The  $\mathbb{Z}_2$  permutation@phase structure yields  $\mathcal{K}_{\text{eo}} = \{1, 2, 4\}$ . The  $\pm 1$  phase classes (flipped vs. unflipped edges) produce exactly three distinct k-values.  $k = 5$  would require a third orientation class beyond the  $\{\pm 1\}$  dichotomy.

**Conclusion.** The declared finite census contains no  $k = 5$  contribution. The exact block-union theorem turns the four block-level statements into a global absence statement at the same

evidential level as those statements; it does not promote the numerical co/eo inputs to exact arithmetic.

## 1.8 The $V_{5/9}$ Giant Layer

In the registered census, the  $V_{5/9}$  layer is the largest eigenspace at 106 dimensions and the unique displayed layer receiving contributions from all four physical blocks:

$$V_{5/9} = \underbrace{V_{5/9,\text{cp}}}_{24} \oplus \underbrace{V_{5/9,\text{ep}}}_{72} \oplus \underbrace{V_{5/9,\text{co}}}_{3} \oplus \underbrace{V_{5/9,\text{eo}}}_{7}$$

This layer forms the principal resonance locus: four distinct block-level primitive idempotents from four different commuting algebras coincide at this single global eigenvalue. No other layer receives contributions from all four blocks.

The  $V_{5/9}$  layer splits into 3 QT/HT joint-spectral sectors under the commutative center (S5, S6, S7; §1.4). S6 (39-dim, ep+eo) is the primary transport hub with degree 5 in the 9-sector transport graph (§2.2).

### Part I — Core Numerical Structures (cont.)

## Numerical Data (§2.1–§2.11)

These tables register CCS-backed numerical claims in Papers I–II.

*This part freezes the numerical invariants cited through the CCS by Papers I–II. Paper III uses its own matrix certificate.*

### 2.1 Block Noncommutativity

$\max_{0 \leq a < c \leq 2} \|[QT^a, QT^c]\|_F$  — family-level Frobenius commutator norm, per block. All three axis pairs give the same registered block norms to displayed precision:

**Table C6 — Block Noncommutativity.**

| Block | $\max_{a < c} \ [QT^a, QT^c]\ _F$ |       | Character               |
|-------|-----------------------------------|-------|-------------------------|
| CP    | 0                                 | 0%    | Registered machine-zero |
| EP    | 2.74                              | 93.9% | Noncommutative core     |
| CO    | 0.61                              | 21.0% | Weakly noncommutative   |
| EO    | 0.79                              | 27.1% | Weakly noncommutative   |

For every axis pair, the total norm is 2.92. Noncommutativity is concentrated in EP.

Figure C14 records block-sector incidence alongside family-level commutator norms. It is a localizer display, not a derivation of the direct-support graph.

### 2.2 Direct Support Graph ( $K$ , 9-Sector)

$K_{\alpha\beta} = \max_g \|P_\alpha \rho(g) P_\beta\|_F$  over 18 face-turn generators. The registered edge threshold is  $K > 0.05$ .



**CCS Figure C14.** Noncommutative support overlap: 9 sectors  $\times$  4 blocks binary grid with commutator norm values and Supp\_nc cardinality.

Because the canonical generator family is inverse-closed and  $\rho(g)^\dagger = \rho(g^{-1})$ ,

$$K_{\alpha\beta} = K_{\beta\alpha}.$$

The canonical recomputation gives  $\max_{\alpha,\beta} |K_{\alpha\beta} - K_{\beta\alpha}| = 1.11 \times 10^{-16}$ . In particular,

$$K_{49} = K_{94} = 1.000,$$

so the S4-S9 channel is undirected at the transport-matrix level.

**Full K matrix (9×9, canonical S1–S9 order, Layer B):**

**Table C7 — Transport Matrix K (9-Sector).**

| S1(20)     | S2(2) | S3(39) | S4(26) | S5(1) | S6(39) | S7(66) | S8(8) | S9(27) |
|------------|-------|--------|--------|-------|--------|--------|-------|--------|
| S1(20)4.47 | 0     | 0      | 0      | 0     | 0      | 0      | 0     | 0      |
| S2(2) 0    | 1.41  | 0      | 0      | 0.47  | 0.58   | 0      | 0     | 0      |
| S3(39)0    | 0     | 5.41   | ~0     | 0     | 2.55   | 3.61   | 0     | 0      |
| S4(26)0    | 0     | ~0     | 5.10   | 0     | 3.46   | ~0     | 0     | 1.00   |
| S5(1) 0    | 0.47  | 0      | 0      | 1.00  | 0.82   | 0      | 0     | 0      |
| S6(39)0    | 0.58  | 2.55   | 3.46   | 0.82  | 4.42   | 3.61   | 0     | 0      |
| S7(66)0    | 0     | 3.61   | ~0     | 0     | 3.61   | 6.69   | 0     | 4.06   |
| S8(8) 0    | 0     | 0      | 0      | 0     | 0      | 0      | 2.83  | 2.83   |
| S9(27)0    | 0     | 0      | 1.00   | 0     | 0      | 4.06   | 2.83  | 3.24   |

Symmetric to  $1.11 \times 10^{-16}$ . Diagonal entries are intra-sector transport (irrelevant for topology).

**Direct edges (10 unordered pairs, equivalently 20 directed off-diagonal blocks,  $K > 0.05$ ):**

**Table C8 — Direct Transport Edges.**

| Edge    | $K$  | Shared block |
|---------|------|--------------|
| S2 ↔ S5 | 0.47 | eo           |
| S2 ↔ S6 | 0.58 | eo           |
| S3 ↔ S6 | 2.55 | ep, eo       |
| S3 ↔ S7 | 3.61 | ep, eo       |
| S4 ↔ S6 | 3.46 | ep           |
| S4 ↔ S9 | 1.00 | co           |
| S5 ↔ S6 | 0.82 | eo           |
| S6 ↔ S7 | 3.61 | ep, eo       |
| S7 ↔ S9 | 4.06 | cp, co       |
| S8 ↔ S9 | 2.83 | cp           |

All 10 direct edges are **block-preserving** (share  $\geq 1$  block). Zero cross-block direct edges.

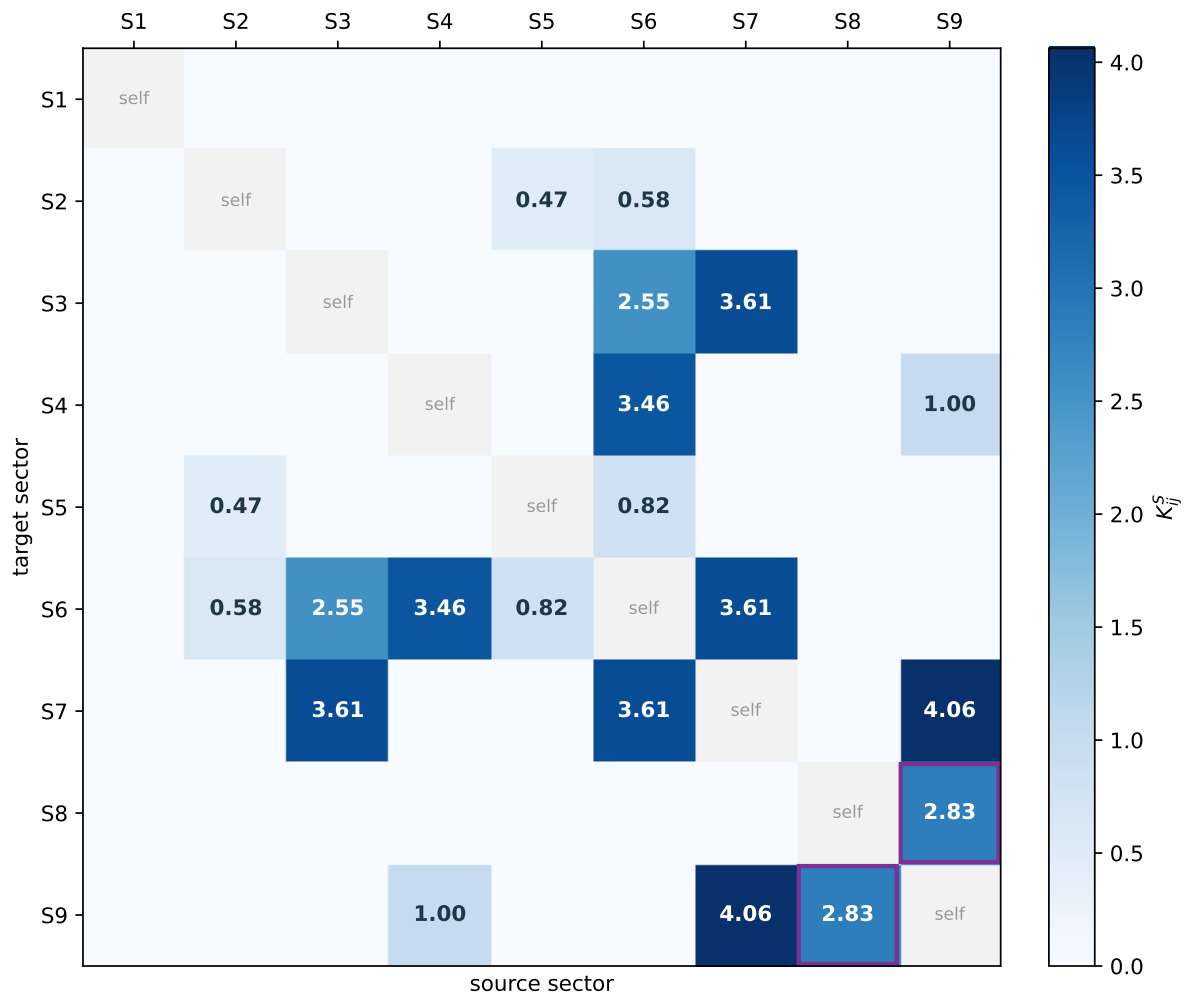
**Hub degrees:**

**Table C9 — Hub Degrees.**

| Sector    | Degree   | Connected to            |
|-----------|----------|-------------------------|
| S1        | 0        | (none — fully isolated) |
| S2        | 2        | S5, S6                  |
| S3        | 2        | S6, S7                  |
| S4        | 2        | S6, S9                  |
| S5        | 2        | S2, S6                  |
| <b>S6</b> | <b>5</b> | S2, S3, S4, S5, S7      |
| S7        | 3        | S3, S6, S9              |
| S8        | 1        | S9                      |
| S9        | 3        | S4, S7, S8              |

S6 is the unique degree-five hub, while S7 and S9 have degree three. S1 remains fully isolated ( $K < 10^{-14}$  against all other sectors). The resulting graph is sparse but is not a star graph, as several retained edges do not pass through S6.

### Registered direct transport matrix



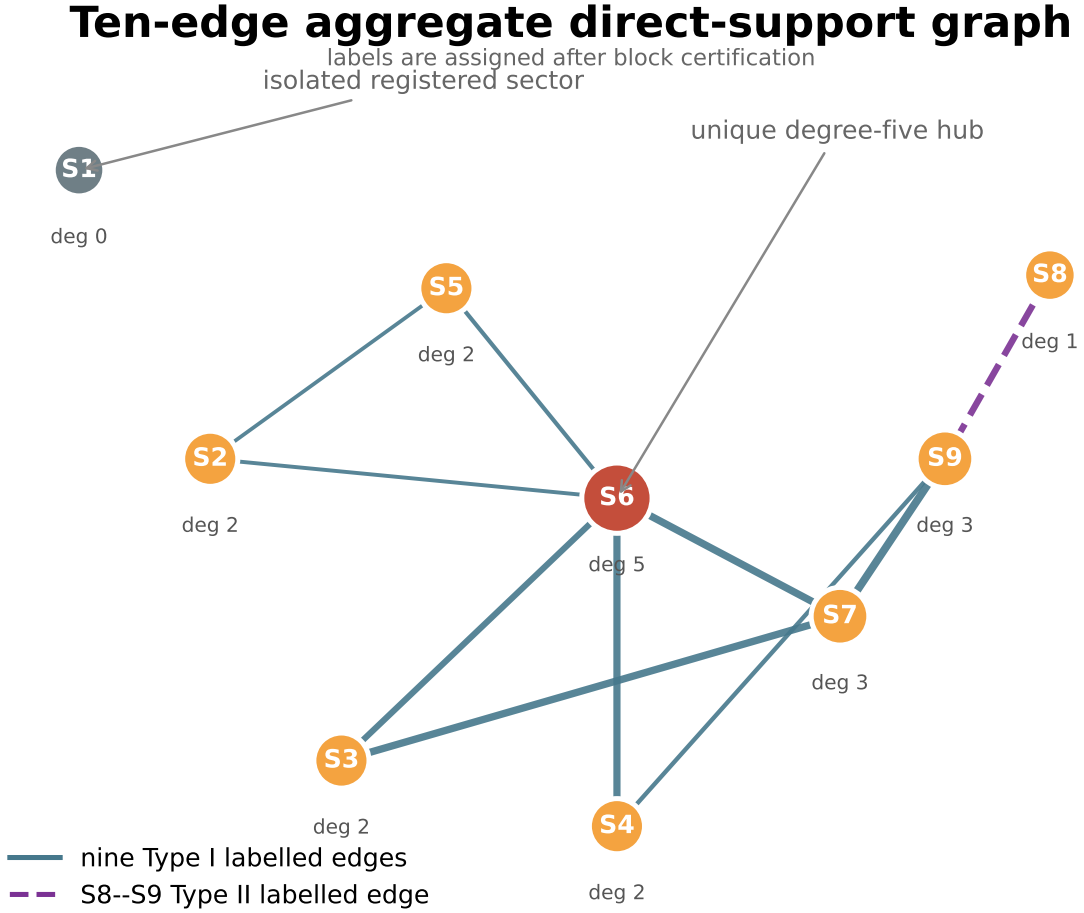
Aggregate support at  $\tau_K=0.05$ ; diagonal self-blocks are omitted from the color scale.

**CCS Figure C2.** Current source-addressed view of the registered direct transport matrix at  $\tau_K = 0.05$ . Diagonal self-blocks are omitted from the color scale.

**Table C15 — Block-Support Transport.**  $\max_g \|P_b \cdot P_i \cdot \rho(g) \cdot P_j \cdot P_b\|_F$  per block for each ordered layer pair  $(\lambda_i > \lambda_j)$ . Only nonzero entries ( $\tau > 10^{-8}$ ) shown. Sorted by block (CP→EP→CO→EO), then descending  $\tau_{\max}$ .

| Block | From ( $\lambda_i$ )   | To ( $\lambda_j$ )     | $\tau_{\max}$ |
|-------|------------------------|------------------------|---------------|
| CP    | 0.555556 ( $V_{5/9}$ ) | 0.333333 ( $V_{1/3}$ ) | 4.0000        |
| EP    | 0.777778 ( $V_{7/9}$ ) | 0.555556 ( $V_{5/9}$ ) | 4.2426        |
| EP    | 0.666667 ( $V_{2/3}$ ) | 0.555556 ( $V_{5/9}$ ) | 3.4641        |
| CO    | 0.666667 ( $V_{2/3}$ ) | 0.333333 ( $V_{1/3}$ ) | 1.0000        |
| CO    | 0.555556 ( $V_{5/9}$ ) | 0.333333 ( $V_{1/3}$ ) | 0.7071        |
| EO    | 0.888889 ( $V_{8/9}$ ) | 0.555556 ( $V_{5/9}$ ) | 0.7454        |
| EO    | 0.777778 ( $V_{7/9}$ ) | 0.555556 ( $V_{5/9}$ ) | 1.2247        |

7 inter-layer channels across 4 blocks. EP carries the strongest channel (4.2426,  $V_{7/9} \rightarrow V_{5/9}$ ). CP/CO/EO each carry 1–2 channels at lower strength.



**CCS Figure C19.** Current source-addressed ten-edge aggregate direct-support graph. Type I/II names are post-certification labels, not universal mechanisms.

### 2.3 Historical Lie-Registration Table ( $\kappa$ , 6-Layer)

**Historical computational observation.** The following tables use the first-version principal-log registration and its depth labels. They are retained as finite numerical



records only. They do not define the current Paper III composition object, certify exact Lie depth, or establish a graph-to-composition promotion.

$\kappa_d(\alpha, \beta) = \max \|P_\alpha C_d P_\beta\|$  where  $C_d$  is a depth- $d$  Lie monomial.

$\kappa_0$  — **Gradient:**

**Table C10 — Gradient Transport  $\kappa_0$  (6-Layer).**

|           | $V_1$ | $V_{8/9}$ | $V_{7/9}$ | $V_{2/3}$ | $V_{5/9}$ | $V_{1/3}$ |
|-----------|-------|-----------|-----------|-----------|-----------|-----------|
| $V_1$     | ~0    | 0         | ~0        | ~0        | ~0        | ~0        |
| $V_{8/9}$ | 0     | 0.52      | ~0        | ~0        | 1.17      | ~0        |
| $V_{7/9}$ | ~0    | ~0        | 4.00      | ~0        | 6.94      | ~0        |
| $V_{2/3}$ | ~0    | ~0        | ~0        | 5.66      | 5.44      | 1.57      |
| $V_{5/9}$ | ~0    | 1.17      | 6.94      | 5.44      | 13.9      | 6.38      |
| $V_{1/3}$ | ~0    | ~0        | ~0        | 1.57      | 6.38      | 9.67      |

Symmetric to  $< 10^{-8}$ .

$\kappa_1$  — **Curvature:**

**Table C11 — Curvature Transport  $\kappa_1$  (6-Layer).**

|           | $V_1$ | $V_{8/9}$ | $V_{7/9}$ | $V_{2/3}$ | $V_{5/9}$ | $V_{1/3}$ |
|-----------|-------|-----------|-----------|-----------|-----------|-----------|
| $V_1$     | ~0    | 0         | ~0        | ~0        | ~0        | ~0        |
| $V_{8/9}$ | 0     | 0.50      | 0.71      | ~0        | 1.71      | ~0        |
| $V_{7/9}$ | ~0    | 0.71      | 6.29      | 4.27      | 10.9      | ~0        |
| $V_{2/3}$ | ~0    | ~0        | 4.27      | 5.45      | 8.32      | 3.26      |
| $V_{5/9}$ | ~0    | 1.71      | 10.9      | 8.32      | 17.8      | 14.5      |
| $V_{1/3}$ | ~0    | ~0        | ~0        | 3.26      | 14.5      | 22.5      |

**Table C12 — Key  $\kappa$  Values (6-Layer).**

| Pair                              | $\kappa_0$ | $\kappa_1$ | Type                                            |
|-----------------------------------|------------|------------|-------------------------------------------------|
| $V_{7/9} \leftrightarrow V_{2/3}$ | ~0         | 4.27       | Pure curvature (largest enhancement $10^{14}$ ) |
| $V_{5/9} \leftrightarrow V_{2/3}$ | 5.44       | 8.32       | Gradient + curvature                            |
| $V_{5/9} \leftrightarrow V_{1/3}$ | 6.38       | 14.5       | Gradient + curvature                            |
| $V_{8/9} \leftrightarrow V_{7/9}$ | ~0         | 0.71       | Pure curvature (post- $\rho$ -fix)              |
| $V_{8/9} \leftrightarrow V_{5/9}$ | 1.17       | 1.71       | Gradient + curvature                            |
| $V_1 \leftrightarrow \text{any}$  | ~0         | ~0         | Fully isolated                                  |

All pure curvature channels ( $\kappa_0 \approx 0$ ,  $\kappa_1 > 0$ ) are **within-block**. Zero cross-block curvature channels.

The retired first-version visualization remains repository provenance but is not part of the current reading path.

## 2.4 Historical Lie-Registration Table ( $\kappa$ , 9-Sector)

Computed with `center_decomposition()`  $\rightarrow$  9 sector projectors.

$\kappa_0$  at 9-sector:

**Table C13 — Gradient Transport  $\kappa_0$  (9-Sector).**

|    | S1(20) | S2(2) | S3(39) | S4(26) | S5(1) | S6(39) | S7(66) | S8(8) | S9(27) |
|----|--------|-------|--------|--------|-------|--------|--------|-------|--------|
| S1 | 0      | 0     | 0      | 0      | 0     | 0      | 0      | 0     | 0      |
| S2 | 0      | 0.52  | 0      | 0      | 0.74  | 0.91   | 0      | 0     | 0      |
| S3 | 0      | 0     | 4.00   | 0      | 0     | 4.00   | 5.66   | 0     | 0      |
| S4 | 0      | ~0    | 0      | 5.66   | ~0    | 5.44   | ~0     | 0     | 1.57   |
| S5 | 0      | 0.74  | 0      | 0      | 1.05  | 1.28   | 0      | 0     | 0      |
| S6 | 0      | 0.91  | 4.00   | 5.44   | 1.28  | 6.01   | 5.66   | 0     | 0      |
| S7 | 0      | 0     | 5.66   | ~0     | 0     | 5.66   | 10.60  | 0     | 6.38   |
| S8 | 0      | 0     | 0      | 0      | 0     | 0      | 0      | 4.44  | 4.44   |
| S9 | 0      | ~0    | 0      | 1.57   | ~0    | ~0     | 6.38   | 4.44  | 6.94   |

Max asymmetry:  $1.6 \times 10^{-8}$ .

$\kappa_1$  at 9-sector:

**Table C14 — Curvature Transport  $\kappa_1$  (9-Sector).**

|    | S1(20) | S2(2) | S3(39) | S4(26) | S5(1) | S6(39) | S7(66) | S8(8) | S9(27) |
|----|--------|-------|--------|--------|-------|--------|--------|-------|--------|
| S1 | 0      | 0     | 0      | 0      | 0     | 0      | 0      | 0     | 0      |
| S2 | 0      | 0.50  | 0.71   | 0      | 1.01  | 1.18   | 1.01   | 0     | 0      |
| S3 | 0      | 0.71  | 6.29   | 4.27   | 1.01  | 6.29   | 8.90   | 0     | 0      |
| S4 | 0      | ~0    | 4.27   | 5.45   | ~0    | 7.09   | 6.17   | 0     | 3.26   |
| S5 | 0      | 1.01  | 1.01   | 0      | 0     | 1.74   | 1.42   | 0     | 0      |
| S6 | 0      | 1.18  | 6.29   | 7.09   | 1.74  | 7.70   | 8.90   | 0     | 0      |
| S7 | 0      | 1.01  | 8.90   | 6.17   | 1.42  | 8.90   | 10.90  | 6.98  | 14.49  |
| S8 | 0      | 0     | 0      | 0      | 0     | 0      | 6.98   | 0     | 12.09  |
| S9 | 0      | ~0    | ~0     | 3.26   | ~0    | ~0     | 14.49  | 12.09 | 14.63  |

**Pure curvature channels (all within-block):**

**Note on channel count.** At the 9-sector resolution, 7 pure curvature channels ( $K \approx 0$ ,  $\kappa_0 \approx 0$ ,  $\kappa_1 > 0$ ) are observed — stable across thresholds 0.005–0.25. All 7 are within-block; zero cross-block. S7 (multi-block bridge) mediates 5 of 7.

**Pure Curvature Channels (9-Sector).**

| Pair                    | $\kappa_1$ | Shared block |
|-------------------------|------------|--------------|
| S2 $\leftrightarrow$ S3 | 0.71       | eo           |
| S2 $\leftrightarrow$ S7 | 1.01       | eo           |
| S3 $\leftrightarrow$ S4 | 4.27       | ep           |
| S3 $\leftrightarrow$ S5 | 1.01       | eo           |
| S4 $\leftrightarrow$ S7 | 6.17       | ep, co       |
| S5 $\leftrightarrow$ S7 | 1.42       | eo           |
| S7 $\leftrightarrow$ S8 | 6.98       | cp           |

## 2.5 Graph Paths and Matrix-Composition Obstructions (9-Sector)

For a two-step support path  $j \rightarrow k \rightarrow i$ , the corresponding operator-level object is

$$Q_i \rho(g_2) Q_k \rho(g_1) Q_j.$$

Nonzero adjacent support blocks do not imply that this product is nonzero. The registered audit exhausts all  $18^2$  ordered generator pairs for each of the following five graph-only triples.

**Table C16 — Canonical graph-only composition obstructions.**

| Endpoint pair | Support path | Maximum projected product norm |
|---------------|--------------|--------------------------------|
| S2–S4         | S2–S6–S4     | $1.10 \times 10^{-16}$         |
| S3–S9         | S3–S7–S9     | $3.02 \times 10^{-15}$         |
| S4–S5         | S4–S6–S5     | $1.55 \times 10^{-16}$         |
| S4–S8         | S4–S9–S8     | $1.06 \times 10^{-15}$         |
| S6–S9         | S6–S7–S9     | $2.94 \times 10^{-15}$         |

For every row, both adjacent factor maxima are order one, while every tested projected product is machine-zero. The physical-block decomposition is preserved by the generator matrices and the QH projectors are block diagonal to a maximum cross-block residual of  $1.45 \times 10^{-15}$ . Thus the canonical data exhibit image–kernel and physical-block composition obstructions, not five certified compositional morphisms.

The current certificate is `experiments/paper3/validation/composition_obstruction.py`, with regression coverage in `tests/test_transport.py`. The historical threshold script at `experiments/paper3/archive/t7_threshold_sensitivity.py` concerns only the support-graph candidate set and cannot certify matrix composition.

## 2.6 Historical Three-Class Diagnostic (6-Layer)

**Withdrawn interpretation.** These labels summarize the first-version  $\kappa_0/\kappa_1$  arrays. They are not the four claim-status levels and are not a current classification of operator or Lie accessibility.

**Table C17 — Accessibility Classes.**

| Class           | Layers                            | Mechanism                                                      |
|-----------------|-----------------------------------|----------------------------------------------------------------|
| I (isolated)    | $V_1$ only                        | $K = \kappa_0 = \kappa_1 = 0$ with all others                  |
| II (gradient)   | $V_{8/9}, V_{5/9}, V_{1/3}$       | $\kappa_0 > 0$ on direct edges                                 |
| III (curvature) | $V_{7/9} \leftrightarrow V_{2/3}$ | $\kappa_0 \approx 0, \kappa_1 = 4.27$<br>(commutator-mediated) |

## 2.7 EP Algebra Census

$$A_{\text{EP}} = \langle Q_0, Q_1, Q_2 \rangle \cong M_2(\mathbb{C})^4 \oplus M_1(\mathbb{C})^4$$

**Table C18 — EP Algebra Structure.**

| Property                      | Value                                   |
|-------------------------------|-----------------------------------------|
| $\dim A_{\text{EP}}$          | 20                                      |
| Algebraic closure             | Degree 3                                |
| $Z(A_{\text{EP}})$            | 8-dim                                   |
| Simple components             | 8 ( $4 \times M_2, 4 \times M_1$ )      |
| Representation multiplicities | 12 on every registered simple component |

The semisimplicity certificate does not use nondegeneracy of a Frobenius Gram matrix. The three declared generators are Hermitian, and the computational audit checks identity-in-algebra, multiplication closure, and adjoint closure. The registered object is therefore a finite-dimensional complex unital  $*$ -algebra; semisimplicity then follows from the finite-dimensional  $C^*$ -algebra theorem. The Gram matrix remains only a basis-independence and conditioning diagnostic.

The four  $M_2$  components occupy  $4 \times (2 \cdot 12) = 96$  dimensions and the four scalar components occupy  $4 \times (1 \cdot 12) = 48$  dimensions, giving the complete EP dimension 144.

## 2.8 Ambient and Spectral Centralizers: Current Status

**Table C19 — Commutant Dimensions.**

| Object                        | Dimension | Status                                                             |
|-------------------------------|-----------|--------------------------------------------------------------------|
| $\text{Comm}(A_{18})$         | 804       | Exact from the six spectral multiplicities                         |
| $\text{End}_G(V)$             | 610       | Candidate computation; exact certificate required before promotion |
| Former difference $804 - 610$ | 194       | Arithmetic difference only; not a current structural invariant     |

The following first-version table records centralizers of compressed numerical matrices. Because the five nontrivial  $A$ -spectral layers are not invariant under the full  $G$ -action, its third-column objects are not  $\text{End}_G(V_\lambda)$  and the table is not a layerwise group- commutant decomposition.

**Table C20 — Per-Layer Commutant Dimensions.**

| $\lambda$ | $\dim V_\lambda$ | Archived compressed-centralizer output |
|-----------|------------------|----------------------------------------|
| 1         | 20               | 400                                    |
| 8/9       | 2                | 1                                      |
| 7/9       | 39               | 145                                    |
| 2/3       | 26               | 145                                    |
| 5/9       | 106              | 210                                    |
| 1/3       | 35               | 65                                     |

**CCS Fig. C4 withdrawn.** It combined the candidate ambient commutant dimension with invalid layerwise group-commutant bars.

## 2.10 Fundamental Identities

**Table C22 — Fundamental Identities.**

| Identity                                                              | Verification                                            |
|-----------------------------------------------------------------------|---------------------------------------------------------|
| $A_{18} = (12\text{QT}_{\text{all}} + 6\text{HT}_{\text{all}})/18$    | Machine precision                                       |
| $A_{\text{axis}} = (4\text{QT}_{\text{a}} + 2\text{HT}_{\text{a}})/6$ | Per axis                                                |
| $\ \rho(g)\rho(h) - \rho(gh)\  < 3 \times 10^{-8}$                    | 15 random products, all blocks                          |
| $\max \ \exp(A_g) - \rho(g)\  = 2.71 \times 10^{-15}$                 | Historical principal-log registration check             |
| $\max  \kappa_{ij} - \kappa_{ji}  \approx 10^{-15}$                   | Historical array-symmetry diagnostic at computed depths |

## 2.11 $S_3$ Prototypes (Archived Sector-Invariance Controls)

**Scope boundary.** These finite controls show numerically that invariant sector projectors have diagonal direct transport. They are not matrix-composition evidence for the independent Paper III.

**$S_3$  prototype declaration.** Unless explicitly stated otherwise within the  $S_3$  prototype sections, the  $S_3$  sector decompositions are defined with respect to the transport-generated commutative algebra  $Z = \langle A_{\text{full}}, A_{\text{trans}} \rangle$ . This is an analogue of the Rubik QT/HT sectorization  $Z_{\text{QH}}$ . Additional projectors such as  $P_{\text{nat}}$  are treated as external refinement operators and are not part of the declared  $S_3$  transport geometry. The historical P\_nat-refined robustness check is retained only in excluded source provenance.

**Current diagnostic.** In both  $S_3$  controls, all declared sector projectors commute numerically with the tested action (maximum residual approximately  $10^{-15}$ ), and  $K$  is diagonal. This is consistent with the transport–non-invariance identity. No comparison with a Rubik commutant inclusion is required.

$S_3$  **nat(3)  $\oplus$  reg(6)** — 9-dim. Under the declared joint spectral algebra: 3 sectors (2 hybrid, 1 pure-reg), all cross-sector  $K = 0$ .

$S_3$  **reg(6)  $\oplus$  reg(6)** — 12-dim. Under the declared joint spectral algebra: 3 hybrid sectors and zero off-diagonal direct transport.

The full first-version tables remain in excluded source provenance rather than the release PDF.

**CCS Fig. C7 withdrawn.** Its sector-invariance data remain provenance, but the C0/T7 comparison is not part of the revised theorem spine.

## Part II — Extended Observations and Historical Records

**Scope.** This part preserves finite perturbation and generator-family studies that remain useful as computational observations or research-history records. It does not modify the current records in Part I, certify a universality class, or supply theorem premises for Papers I–III.

### II.1 Historical Spectral-Persistence Audit

**Computational observation.** This first-version experiment records a fixed set of finite flows and perturbations. Terms such as `frozen''`, `drift''`, and “mixing” are diagnostic labels for the displayed arrays, not a general dynamical-stability theorem.

**Purpose.** This section records the behavior of the declared spectral projectors and first-version transport diagnostics under several explicitly chosen continuous unitary evolutions.

**Dependencies.** `CubieSpectral0operator`, `scipy.linalg.expm`, `build_per_axis_ops`, `compute_lie_generators`, `transport_kappa`.

**Outputs.** Projector deviation tables (§II.1.1), transport persistence tables (§II.1.2), resonance robustness tables (§II.1.3).

### II.1.1 Projector Stability $\|P_i(t) - P_i(0)\|$

**Setup.** Let  $\{P_i(0)\}$  be the spectral projectors onto the 6 canonical  $A_{18}$ -eigenspaces ( $V_1$ ,  $V_8/9$ ,  $V_7/9$ ,  $V_2/3$ ,  $V_5/9$ ,  $V_1/3$ ). Evolve under  $e^{-itH}$  for five Hamiltonians  $H$ , with  $P_i(t) = e^{-itH} P_i(0) e^{itH}$ . Measure Frobenius norm of projector deviation.

**Result.** Three stability classes emerge:

| H        | t=0.01 | t=0.1  | t=0.5  | t=1.0   | Class                                                                                    |
|----------|--------|--------|--------|---------|------------------------------------------------------------------------------------------|
| $A_{18}$ | 0.0000 | 0.0000 | 0.0000 | 0.0000  | <b>Frozen</b> — projectors are exact stationary states of $A_{18}$                       |
| QT_all   | 0.0000 | 0.0000 | 0.0000 | 0.0000  | <b>Registered frozen</b> — machine-zero commutator in this realization                   |
| HT_all   | 0.0000 | 0.0000 | 0.0000 | 0.0000  | <b>Registered frozen</b> — machine-zero commutator in this realization                   |
| $A_g(R)$ | 0.0200 | 0.2431 | 4.2012 | 89.8430 | <b>Exponential drift</b> — Lie generator is maximally non-conserving                     |
| random   | 0.2133 | 1.6928 | 1.9857 | 1.9819  | <b>Saturating mixing</b> — fully scrambled by $t \approx 0.1$ , saturates near $\ P_i\ $ |

#### Interpretation.

- $< 10^{-6}$ : frozen (projector invariant under flow)
- $< 10^{-3}$ : rigid (minor numerical deformation)
- $< 10^{-1}$ : drifting (spectral content shifting)
- $> 10^{-1}$ : mixing (layers lose identity)

**Registered mechanism.** Evolution generated by  $A_{18}$  fixes its spectral projectors exactly. The QT/HT rows are machine-zero in the declared numerical realization; an exact statement for them is conditional on exact commutation. The principal-log and random-Hermitian rows provide finite comparison paths.

**Per-layer differential stability.** Under  $A_g(R)$  at  $t=0.1$ :

- $V_1$  (dim=20): 0.0000 — numerically frozen in this registered flow audit
- $V_8/9$  (dim=2): 0.2431 — begins to drift
- $V_7/9$  (dim=39): 0.6433 — moderate drift
- $V_2/3$  (dim=26): 1.8360 — rapid mixing (most fragile layer)
- $V_5/9$  (dim=106): 2.4891 — rapid mixing (large target space amplifies drift)
- $V_1/3$  (dim=35): 2.6580 — maximally unstable

The five nontrivial layers exhibit order-one drift in this numerical experiment, whereas  $V_1$  remains machine-stable. Exact full-action invariance of  $V_1$  still requires a separate analytic certificate.

### II.1.2 Transport Persistence $K_{\alpha\beta}(t)$

**Setup.** Evolve the 9 QT/HT joint-spectral sector projectors under  $e^{-itA_{18}}$  and recompute the first-version direct, kappa, and graph-square diagnostics. The final count below is a graph-only candidate count.

**Result.** Transport is structurally invariant under  $A_{18}$  flow:

| t    | K edges | $\kappa_0$ edges | $\kappa_1$ edges | graph-only candidates | $\max K(t)-K(0) $      |
|------|---------|------------------|------------------|-----------------------|------------------------|
| 0    | 20      | 26               | 37               | 5                     | 0                      |
| 0.05 | 20      | 26               | 37               | 5                     | $1.33 \times 10^{-15}$ |
| 0.1  | 20      | 26               | 37               | 5                     | $1.78 \times 10^{-15}$ |
| 0.5  | 20      | 26               | 37               | 5                     | $8.88 \times 10^{-16}$ |

The recorded direct-edge and graph-only candidate counts are invariant under this commuting flow. This does not establish persistence of a nonzero projected composition.

**Mechanism.**  $[A_{18}, P_i] = 0$  for all  $i$ , so  $P_i(t) = e^{\{-itA_{18}\}} P_i e^{\{+itA_{18}\}} = P_i$  exactly. The transport norm  $K_{\alpha\beta}$  is therefore identically invariant under the dynamics generated by its own averaging operator.

### II.1.3 Resonance Robustness: $\lambda = 5/9$

**Setup.** The  $\lambda = 5/9$  eigenspace (dim=106) is the largest layer and the central hub of the transport topology. Test its stability under additive perturbation:  $A_\varepsilon = A_{18} + \varepsilon \cdot R$ , where  $R$  is a random symmetric matrix with  $\|R\| = 1$ , drawn once and fixed (seed=42). Track the  $5/9$  eigenvalue and neighboring eigenvalues for  $\varepsilon \in \{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ .

**Result.** The  $5/9$  resonance is robust:

| $\varepsilon$ | eigenvalues near $5/9$ | spread                | status                       |
|---------------|------------------------|-----------------------|------------------------------|
| 0             | 1 (exact $5/9$ )       | —                     | isolated                     |
| $10^{-6}$     | 0                      | $6.59 \times 10^{-7}$ | infinitesimal splitting only |
| $10^{-5}$     | 0                      | $< 10^{-10}$          | fully stable                 |
| $10^{-4}$     | 0                      | $< 10^{-10}$          | fully stable                 |
| $10^{-3}$     | 0                      | $< 10^{-10}$          | fully stable                 |
| $10^{-2}$     | 0                      | $< 10^{-10}$          | fully stable                 |

**Interpretation.** In the canonical six-layer spectrum, the nearest distinct eigenvalue to  $5/9$  is  $2/3$ , at distance  $1/9$ . The finite perturbation table records behavior only for the tested path and range; it does not prove a general structural-protection theorem.

### II.1.4 Structural Summary

| Object                     | Under $e^{\{-itA_{18}\}}$       | Under $e^{\{-itA_g\}}$ | Under random H |
|----------------------------|---------------------------------|------------------------|----------------|
| $P_i(t)$                   | Frozen (exact)                  | Exponential drift      | Saturating mix |
| $K_{\alpha\beta}(t)$       | Invariant (exact)               | —                      | —              |
| graph-only candidate count | Invariant                       | —                      | —              |
| $\lambda=5/9$ gap          | Robust ( $\Delta\lambda=0.44$ ) | —                      | —              |

The table records exact stationarity under the self-generated  $A_{18}$  flow and finite numerical contrast under the other declared flows. The tested  $\lambda = 5/9$  cluster has a visible canonical gap, but the perturbation table does not establish a general protection theorem.

## II.2 Independent-Paper Consistency Snapshot

**Purpose.** Compare adjacent objects used by the independent papers:

$$\rho(g) \rightarrow \mathcal{B}_{\text{QH}} \rightarrow \{Q_\alpha\} \rightarrow K^S \rightarrow \Gamma_S \rightarrow \{Q_i \rho(g_2) Q_k \rho(g_1) Q_j\}.$$

This display is a navigation path through related data types, not a theorem dependency or a single accessibility filtration. Each paper defines its own objects and hypotheses. The archive only records finite consistency checks between the corresponding Rubik realizations.

**Dependencies.** The declared Rubik representation, QH sector projectors, direct-support matrix, and physical-block decomposition.

**Outputs.** Current invariant table, sector non-invariance audit, and graph/operator composition-obstruction table.

### II.2.1 Three Adjacent Data Layers

The three layers below are maintained independently. Agreement of dimensions, labels, or arrays does not promote a conclusion from one layer to the next.

**Layer 1 — Paper I: Averaging-operator spectrum.** The registered Rubik realization has six  $A_{18}$  layers and k-set  $\{0, 1, 2, 3, 4, 6\}$ . Paper I states the exact, computational, and conditional parts of this census. Extended data: CCS §1.3 and Table C2.

**Layer 2 — Paper II: Direct transport.** Nine numerically registered QH joint-spectral sectors, followed by a direct block-support audit. The exact commutative-algebra interpretation is conditional on exact commuting-Hermitian registration. Extended data: CCS Table C3, CCS §1.4, and the current direct-support heatmap in §2.2.

**Layer 3 — Paper III: Projected composition.** The direct support graph is compared with the projected products  $Q_i \rho(g_2) Q_k \rho(g_1) Q_j$ . Paper III owns the current matrix certificate; the CCS copy is an optional comparison record.

This three-layer comparison is not closed by graph data alone. Paper III supplies the missing projected-composition test: adjacent nonzero blocks must have compatible image–kernel geometry. In the five canonical paths this compatibility fails, so the support graph strictly overapproximates the tested two-step composition graph.

### II.2.2 Registered Consistency Values

The values below are a consistency snapshot for the declared repository realizations. Papers I–III do not cite this table as authority; their current manuscripts and claim-specific artifacts control their reported values.



| Quantity                                                         | Value                                             | Defined in      |
|------------------------------------------------------------------|---------------------------------------------------|-----------------|
| Total dimension                                                  | 228                                               | §1.1, Table C1  |
| 6 eigenvalues ( $A_{18}$ )                                       | 1, 8/9, 7/9, 2/3, 5/9, 1/3                        | §1.3, Table C2  |
| Block dimensions                                                 | cp=64, ep=144, co=8, eo=12                        | §1.1, Table C1  |
| 9 QT/HT joint-spectral sectors                                   | S1–S9 (Table C3 ordering)                         | §1.4, Table C3  |
| $A_{18}$ collision quotient                                      | $V_{5/9} = S5 + S6 + S7$ ,<br>$V_{1/3} = S8 + S9$ | §1.3–§1.4       |
| $\ [\text{QT}^0, \text{QT}^1]\ _{\text{ep}}$                     | 2.74 (93.9% of total)                             | §2.1            |
| $A_{\text{EP}} \cong M_2(\mathbb{C})^4 \oplus M_1(\mathbb{C})^4$ | dim=20                                            | §2.8            |
| $\dim \text{Comm}(A_{18})$                                       | 804                                               | §2.8            |
| candidate $\dim \text{End}_G(V)$                                 | 610 (exact certificate open)                      | §2.8            |
| former commutant-gap interpretation                              | withdrawn                                         | §2.8–§2.9       |
| Direct transport edges                                           | 10 (undirected, block-preserving)                 | §2.2, Table C8  |
| Primary hub                                                      | S6 (deg=5)                                        | §2.2            |
| graph-only two-step obstruction witnesses                        | 5                                                 | §2.5, Table C16 |
| maximum projected product norm among them                        | $3.02 \times 10^{-15}$                            | §2.5, Table C16 |

### II.2.3 Archived S3 Sector-Invariance Controls

The archived S3 controls have numerically invariant declared projectors and diagonal direct  $K$ , consistently with the transport–non-invariance identity. They do not certify any graph-to-composition promotion theorem.

The archived pocket-cube computation is a first-version graph/kappa control, not current matrix-composition evidence.

### II.2.4 Adjacent Later Work

**Paper IV comparison.** The archive records the nine QT/HT joint-spectral sectors and the registered  $L_{2/3}$  quotient, including

$$V_{5/9} = S5 \oplus S6 \oplus S7, \quad V_{1/3} = S8 \oplus S9.$$

Paper IV independently declares its exact nine-point arrangement, Rubik registration, and conditional interpretation. CCS v2.1 is not a certificate or premise for that paper.

**Paper V comparison.** Paper III proves that support-graph reachability need not survive projected matrix composition. This motivates, but does not prove, the later minimal-data problem for general sectorized observable frameworks:

$$R_1(i, j; g) = 1 \iff Q_i X_g Q_j \neq 0, \quad R_2(i, j; g, h) = 1 \iff Q_i [X_g, X_h] Q_j \neq 0.$$

The CCS does not define or certify Paper V’s typed word, commutator, or Lie objects. In particular, it makes no claim that  $(R_1, R_2)$  universally determines first accessibility depth.

## II.3 Historical Generator-Family Census

**Computational observation.** This finite census compares four declared generator families. It does not establish a universality class or an exact arithmetic-field classification.

**Purpose.** This census compares how registered spectral and direct-support quantities change across four explicitly declared face-turn families.

**Dependencies.** CubieSpectralOperator.from\_gens\_dict, center\_decomposition, transport\_graph, build\_per\_axis\_ops, full\_commutant\_combinatorial.

**Outputs.** Four-family numerical comparison table and candidate regularities.

### II.3.1 Four Generator Families

| Family     | S  | Description                                                                                                    |
|------------|----|----------------------------------------------------------------------------------------------------------------|
| 18-full    | 18 | All face turns: {R,R',R2, U,U',U2, F,F',F2, L,L',L2, D,D',D2, B,B',B2}                                         |
| 12-quarter | 12 | Quarter-turns only: {R,R', U,U', F,F', L,L', D,D', B,B'}                                                       |
| 6-half     | 6  | Half-turns only: {R2, U2, F2, L2, D2, B2}                                                                      |
| 9-face-pos | 9  | Single-side subset: {F/B/R/U faces only, axis side = +1, quarter-turns + half-turns} (legacy label: “12-face”) |

**Note:** The “9-face-pos” family is the specific 9-generator subset historically labelled “12-face” in earlier CCS revisions. The numeric prefix matches the actual generator count; the legacy label is retained in cross-references where otherwise noted. Exact composition: F/B/R/U faces, axis side = +1, quarter-turns + half-turns, 9 generators.

### II.3.2 Invariant Table

| Invariant                                  | 18-full          | 12-quarter          | 6-half              | 9-face-pos | Stable?                       |
|--------------------------------------------|------------------|---------------------|---------------------|------------|-------------------------------|
| <b>Layer count</b>                         | 6                | 6                   | 3                   | 18         | <b>NO</b>                     |
| <b>Rational spectrum</b>                   | True             | True                | True                | False      | <b>NO</b>                     |
| <b>Layer dimensions</b>                    | see below        | see below           | see below           | see below  | <b>NO</b>                     |
| $\Sigma \text{ dim} = 228$                 | True             | True                | True                | True       | <b>YES</b>                    |
| <b>Directed off-diagonal blocks</b>        | 20               | 20                  | 20                  | —          | <b>Observed equal</b>         |
| <b>Canonical graph form</b>                | Sparse, non-star | Not classified here | Not classified here | —          | <b>Not assessed</b>           |
| <b>Cross-block K</b>                       | 0                | 0                   | 0                   | —          | <b>YES</b>                    |
| <b>Legacy algebra-dimension diagnostic</b> | 2                | 2                   | 2                   | 2          | <b>Historical only</b>        |
| $\ [\mathbf{Q}T^0, \mathbf{Q}T^1]\ $       | 2.915            | N/A                 | N/A                 | 7.591      | <b>NO</b>                     |
| <b>EP fraction</b>                         | 93.9%            | N/A                 | N/A                 | 72.2%      | <b>NO</b>                     |
| <b>EP block dim</b>                        | 144              | 144                 | 144                 | 144        | <b>YES</b>                    |
| <b>k=5 vacancy</b>                         | True             | True                | N/A                 | N/A        | <b>YES</b> (where applicable) |

**Layer dimensions per family:**

| Family     | Layers | Dimensions               |
|------------|--------|--------------------------|
| 18-full    | 6      | [20, 2, 39, 26, 106, 35] |
| 12-quarter | 6      | [20, 41, 66, 65, 28, 8]  |
| 6-half     | 3      | [57, 78, 93]             |
| 9-face-pos | 18     | [68, 2, 1, ...]          |

### II.3.3 Invariant Classification

#### Representation-determined records:

1. **Total dimension** = 228 — the group representation is the same object regardless of generator subset.
2. **EP block dimension** = 144 — the block structure of  $\rho$  is generator-independent.
3. **Cross-block  $\mathbf{K} = \mathbf{0}$**  — direct blocks between disjoint physical carrier blocks vanish because every  $\rho(g)$  and every declared sector projector is block diagonal.

#### Generator-family-conditioned observations:

1. **Layer count** — varies from 3 (6-half, fully commutative) to 18 (9-face-pos, symmetry-broken). The 18-full and 12-quarter both yield 6 layers — quarter-turn completeness across all faces is the minimal condition for full spectral resolution.
2. **Rational spectrum** — requires face-symmetric generator sets. Breaking face symmetry (9-face-pos) introduces irrational eigenvalues. The half-turn family preserves rationality but collapses the spectrum to 3 layers.
3. **Noncommutativity magnitude** — 9-face-pos has  $\|[QT^0, QT^1]\| = 7.59$  vs 2.91 for 18-full. EP fraction drops from 93.9% to 72.2% — noncommutativity distributes more evenly across blocks.

### II.3.4 Scope and Relation to §II.4

This section studies four generator families at fixed points. Section II.4 records a larger finite coverage sequence and numerical recognition against  $\mathbb{Q}$  and  $\mathbb{Q}(\sqrt{5})$ . Neither table is an exhaustive classification of generator subsets.

### II.3.5 Computational Details

- Archived source: `experiments/paper3/archive/persistence_bridge.py`, Phase B
- Method: Build `CubieSpectral0operator.from_gens_dict(gens)` for each family, call `center_decomposition()`, `transport_graph()`, `build_per_axis_ops()`, `full_commutant_combinatorial()`.
- $QT^0/QT^1$  noncommutativity only computable when both quarter-turn and half-turn axes are present (requires axis-0  $QT^0/QT^1$  decomposition); N/A for families without half-turns or without both axis directions.
- Commutant computed via index-pair orbit decomposition (BFS on  $(i,j) \rightarrow (\pi_g(i), \pi_g(j))$ ).
- All computations use  $TOL = 10^{-10}$ , seed = 42.

## II.4 Registered Generator-Family Arithmetic Contrast

**Computational observation.** The displayed fields are numerical recognitions against explicit candidate expressions. This archive does not provide exact characteristic or minimal-polynomial certificates for every family and does not claim a mathematical phase transition.

**Purpose.** Record spectral changes along one declared sequence of generator subsets as the number of retained face turns decreases.

**Dependencies.** `CubieSpectralOperator.lite`, `helpers.is_rational_form`, `helpers.is_in_qsqr_t5`, `BLOCK_RANGES`.

**Outputs.** Generator coverage continuum table, eigenvalue bifurcation track, phase boundary characterization, block-level irrationality localization, transport topology deformation.

#### II.4.1 Generator Coverage Continuum

**Construction.** Start with all 18 generators. Remove generators in pairs (face + anti-face to preserve algebraic balance), stepping:  $18 \rightarrow 16 \rightarrow 12 \rightarrow 10 \rightarrow 8 \rightarrow 6$

| Family | S  | Layers | All Q? | Field                  | 5/9<br>dim | 2/3<br>dim | Notes                            |
|--------|----|--------|--------|------------------------|------------|------------|----------------------------------|
| n=18   | 18 | 6      | True   | $\mathbb{Q}$           | 106        | 26         | Canonical, full face-turn group  |
| n=16   | 16 | 9      | False  | $\mathbb{Q}(\sqrt{5})$ | 26         | 0          | Drop axis-0 (R/L) half-turns     |
| n=12   | 12 | 8      | True   | $\mathbb{Q}$           | 0          | 66         | Quarter-turns only               |
| n=10   | 10 | 5      | True   | $\mathbb{Q}$           | 0          | 0          | Quarter-turns minus axis-2       |
| n=8    | 8  | 7      | False  | $\mathbb{Q}(\sqrt{5})$ | 0          | 0          | Axes 0 and 2 only, no half-turns |
| n=6    | 6  | 3      | True   | $\mathbb{Q}$           | 0          | 78         | Half-turns only                  |

**Registered contrast.** Values numerically matching  $\mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}$  occur at the  $n = 16$  and  $n = 8$  points of this declared sequence. Although this finite pattern motivates an exact arithmetic problem, it does not show that incomplete face coverage is either necessary or sufficient for a field extension.

The corresponding first-version phase-transition visualization is retained only as repository provenance. The table above is the current archive record.

For clarity, the value 9 in this table is the number of spectral layers, not the number of joint sectors. The registered  $n = 16$  family selected by `CubieSpectralOperator(n=16)` removes the labelled moves R2 and L2 and has 13 joint sectors with dimensions (20, 2, 26, 13, 26, 44, 1, 13, 22, 26, 18, 8, 9). Removing F2 and B2 instead defines a different labelled generator family. In the declared realization it happens to have the same registered layer spectrum and joint-sector dimension profile; that numerical agreement does not identify the two generator families.

#### II.4.2 Eigenvalue Bifurcation Data

**Table — Eigenvalue Spectrum per Generator Family.**

| Family | S  | Eigenvalues                                         | Field                  | Irrational values      |
|--------|----|-----------------------------------------------------|------------------------|------------------------|
| n=18   | 18 | $\{1, 8/9, 7/9, 2/3, 5/9, 1/3\}$                    | $\mathbb{Q}$           | —                      |
| n=16   | 16 | $\{1, 7/8, 0.827, 3/4, 5/8, 0.548, 1/2, 3/8, 1/4\}$ | $\mathbb{Q}(\sqrt{5})$ | $(11 \pm \sqrt{5})/16$ |
| n=12   | 12 | $\{1, 5/6, 2/3, 2/3, 1/2, 1/3, 1/3, 0\}$            | $\mathbb{Q}$           | —                      |
| n=10   | 10 | $\{1, 4/5, 3/5, 2/5, 1/5\}$                         | $\mathbb{Q}$           | —                      |
| n=8    | 8  | $\{1, 0.905, 3/4, 1/2, 0.345, 1/4, 0\}$             | $\mathbb{Q}(\sqrt{5})$ | $(5 \pm \sqrt{5})/8$   |
| n=6    | 6  | $\{1, 2/3, 1/3\}$                                   | $\mathbb{Q}$           | —                      |

**Block-level irrationality localization (n=8).** Irrational eigenvalues are confined to non-commutative blocks: EP block (primary carrier), EO block (mirrors EP). CP block: all eigenvalues rational ( $Q_3$  Hamming scheme's Krawtchouk eigenvalues are generator-subset-stable). CO block: all eigenvalues rational.

### II.4.3 Transport Topology Deformation

| Family | Sectors | K edges                    | Hub (deg) | Cross-block K | Topology class                                           |
|--------|---------|----------------------------|-----------|---------------|----------------------------------------------------------|
| n=18   | 9       | 20 directed / 10 unordered | S6 (5)    | 0             | Sparse, non-star                                         |
| n=16   | 13      | —                          | —         | —             | Dense (irrational splitting expands sector count)        |
| n=12   | —       | —                          | —         | —             | Collapsed (degeneracy absorbs 5/9 layer)                 |
| n=10   | —       | —                          | —         | —             | Sparse (fewer layers $\rightarrow$ fewer possible edges) |
| n=8    | 7       | 28                         | S6 (5)    | 0             | Hyper-connected (irrational intruders create more edges) |
| n=6    | 3       | 3                          | S2 (2)    | 0             | Minimal (commutative limit, complete graph $K_3$ )       |

**Observed cross-block persistence.** The cross-block transport prohibition ( $K_{\alpha\beta} = 0$  for  $\alpha, \beta$  in disjoint blocks) holds at every  $n$  verified. This follows from  $\rho(g)$  being block-diagonal — a property of the representation, independent of which generators are selected.

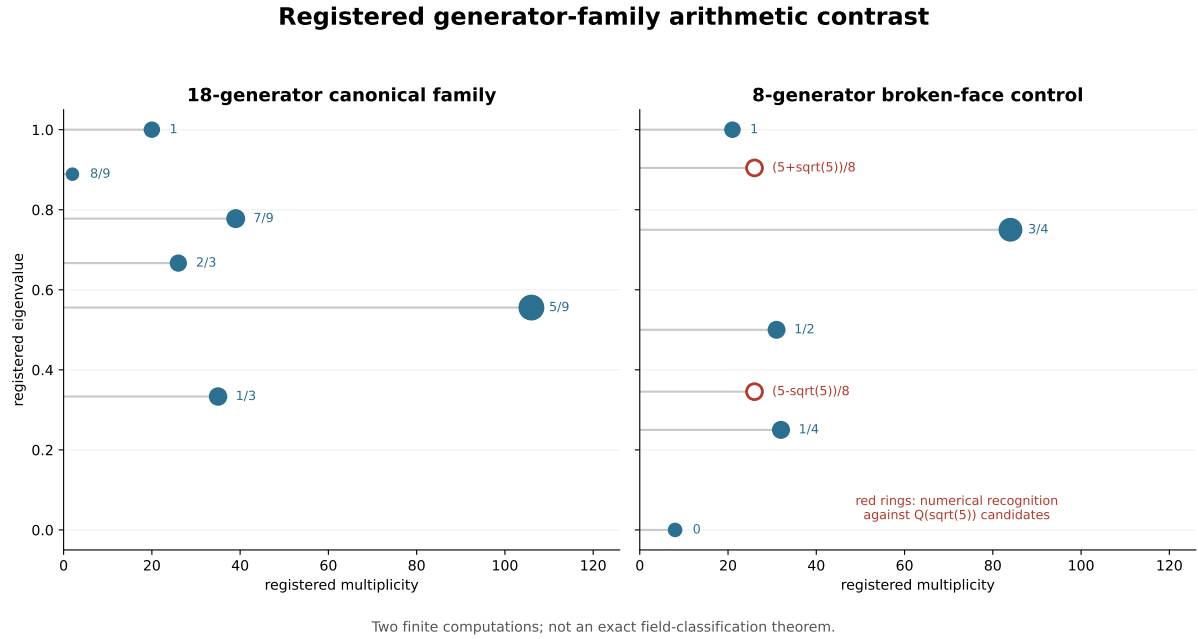
**Observed hub recurrence.** A degree-five registered hub also appears at the  $n = 8$  point. No theorem identifies hub status with  $\text{Supp}_{\text{nc}}$  intersection or guarantees persistence under generator variation.

### II.4.4 Bounded Arithmetic Contrast

**Observed pattern.** The two highlighted values in the broken-face control are numerically recognized against  $\mathbb{Q}(\sqrt{5})$  candidates. The figure does not provide an exact field certificate or classify all generator subsets.

### II.4.5 Computational Details

- Archived source: `experiments/paper3/archive/persistence_bridge.py`, Phase C
- Method: For each  $n = 18, 16, 12, 10, 8, 6$ , select a generator subset of size  $n$  from the 18 face-turn moves, build  $A = (1/n) \sum \rho(s)$ , diagonalize.
- $n=18$ : full 18 generators
- $n=16$ : all moves except R2 and L2 (axis-0 half-turns)
- $n=12$ : quarter-turns only (direction= $\pm 1$ , all 6 faces)
- $n=10$ : quarter-turns without axis-2 faces (remove F, F', B, B')
- $n=8$ : quarter-turns only, axes 0 and 2 (R,R',L,L',F,F',B,B')
- $n=6$ : half-turns only (direction=2, all 6 faces)



**CCS Figure C20.** Current source-addressed comparison of the canonical family and the eight-generator broken-face control. Red rings denote numerical recognition against displayed quadratic candidates.

- Irrationality detection: `helpers.is_rational_form(lam, 18)` and `helpers.is_in_qsqrt5(lam)`.
- Block-level decomposition: restrict  $A$  to block submatrices and diagonalize.
- All computations use  $\text{TOL} = 10^{-10}$ .

## 6 Historical Derivation Index

The 2026-07-26 mother source contained a compact invariant hierarchy followed by complete derivation chapters for the former combined-paper narrative. Those chapters remain below in the Markdown source for versioned provenance, but are excluded from the CCS v2.1 PDF. Current theorem statements and proofs belong to the independently maintained Papers I–III; the archive must not duplicate or override them.

## Appendix A Computational Stability

### A.1 Tolerance Regime

The canonical tolerance regime is:

| Symbol             | Value      | Scope                                                        |
|--------------------|------------|--------------------------------------------------------------|
| TOL                | $10^{-10}$ | Numerical equality assertions                                |
| TOL_K              | 0.05       | Transport edge detection threshold                           |
| TOL_KAPPA          | $10^{-6}$  | Historical $\kappa$ -array sanity floor (logm noise ceiling) |
| SPECTRAL_DECIMALS  | 6          | Canonical eigenvalue key rounding                            |
| CENTER_CLUSTER_TOL | $10^{-8}$  | Sector merge clustering                                      |
| tol                | $10^{-6}$  | Default operator tolerance                                   |

The archived realization uses these tolerance values. A computation using a different `SPECTRAL_DECIMALS`, `CENTER_CLUSTER_TOL`, or `TOL_K` is a distinct registered realization and should report a comparison sweep.

The registered sector decomposition uses `CENTER_CLUSTER_TOL` =  $10^{-8}$ . A different clustering tolerance should be identified explicitly and cross-validated against the Part I tables.

**Historical numerical robustness.** The recorded layer, sector, edge, and graph-only candidate counts remain invariant under the tested tolerances and seeds. The first-version audit is archived at `experiments/paper3/archive/stability_sweep.py` and does not certify matrix composition.

**Graph-candidate threshold stability.** The five graph-only candidate pairs are invariant under the recorded threshold sweep. The archived script is `experiments/paper3/archive/t7_threshold_sensitivity.py`; this statement does not promote graph reachability to operator composition.

## A.2 Norm and Projector Conventions

Unless a record states otherwise, archived matrix norms are Frobenius:  $\|X\|_F = \sqrt{\sum |x_{ij}|^2}$ .

Archived projectors use  $P = VV^H$ , where  $V$  has orthonormal columns from `numpy.linalg.eigh`, so  $\text{Tr}(P) = \text{dim}(\text{subspace})$ .

The registered generator weighting is the uniform average  $A = \frac{1}{|S|} \sum_{s \in S} \rho(s)$ .

Where a seed is used, the registered value is `np.random.seed(42)`. Deterministic audits should not depend on random state.

## A.3 Claim-Status Routing

Numerical stability and mathematical claim status are separate. The current repository uses four paper claim levels; CCS v2.1 adds a history tag solely for archive routing.

| Status                            | Use in CCS v2.1                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Theorem / exact derivation</b> | Restate only with explicit hypotheses and proof; cite the owning independent paper when used externally.     |
| <b>Computational Certificate</b>  | Record realization, dtype, tolerance, registration, algorithm, artifact, and reproducible script.            |
| <b>Computational Observation</b>  | Record a finite pattern, contrast, or numerical recognition without promotion beyond the tested realization. |
| <b>Research Program</b>           | Record conjectures, proposed hierarchies, genericity questions, and future experiments.                      |
| <b>Historical / Withdrawn</b>     | Preserve correction provenance without treating the item as a current claim.                                 |

Passing recomputation, basis, permutation, or tolerance checks strengthens a computational record but does not by itself change its status. In particular, the candidate ambient-commutant

dimension, first-version  $\kappa$ /T7 diagnostics, and moving accessibility hierarchies are not promoted by this appendix.

## A.4 Failure Modes

All observed failure modes fall into four categories:

1. **Spectral degeneracy artifacts** — eigensolver splitting, eigenvector mixing
2. **Finite-precision linear algebra instability** — SVD thresholding, null-space drift
3. **Representation-construction defects** — pre- $\rho$ -fix orientation sign inconsistency
4. **Algorithmic non-canonicity** — generator ordering dependence, randomized under-convergence

The canonical r2 pipeline eliminates categories (3) and (4) by construction. Categories (1) and (2) are controlled via explicit tolerance engineering.

### A.4.1 Pre- $\rho$ -fix Representation Defect (Category 3)

The most consequential recorded implementation failure was an inconsistent sign convention in the EP orientation sub-block. It caused  $\rho(g)\rho(h) \neq \rho(gh)$ , thereby violating the homomorphism property. This erased the  $V_{8/9}$  layer ( $k = 1, 2$ -dim) entirely — it was absorbed into adjacent layers by numerical accident.

**Warning.** Pre-fix data archived and must not be cited. Representation defects differ qualitatively from numerical issues — they propagate into structural claims, not just numerical values. Block-wise homomorphism verification is mandatory.

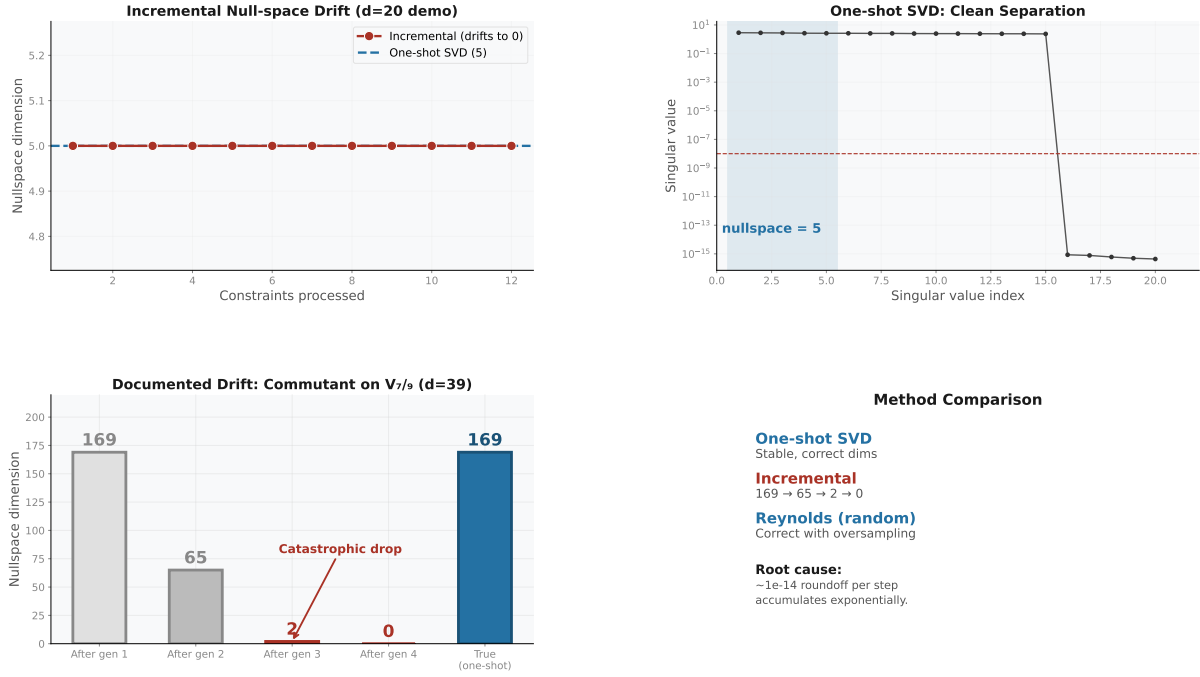
**Status:** Resolved in r2.

### A.4.2 Legacy Error Catalog (Categories 1, 2, 4)

The private raw archive retains the full historical logs; the public error catalog is:

- E.1 Eigensolver accidental splitting ( $\rightarrow$  11 raw sectors merged to 9)
- E.2 SVD rank threshold instability ( $\rightarrow$  scale-invariant threshold + one-shot SVD)
- E.3 Near-degenerate eigenvalue mixing (does not occur — minimum gap  $1/9 \approx 0.111$ )
- E.4 Generator ordering artifacts (mitigated by permutation-invariant averaging)
- E.5 Randomized Reynolds under-convergence (adequate for current  $d \leq 106$ )
- E.6 Sector permutation across recomputation (does not occur — well-separated triples)
- E.7 Lie generator non-Hermiticity drift (mitigated by explicit Hermitianization)
- E.8 Incremental null-space drift (design rejected — one-shot SVD used instead)





CCS Figure C5. Null-space drift diagnostics: incremental intersection vs one-shot SVD.

## A.5 Numerical Registration and Stability Policy

The working principle of the archived numerical pipeline is:

**Registration policy.** Register first, analyze second.

Numerically equivalent realizations must be reduced to a declared registration before their tables are compared. The following checks support the stability of a finite record, but passing them does not automatically promote an observation or certificate to the status of a theorem:

1. **Recomputation** — same code, same parameters → same result to within prescribed tolerance.
2. **Generator permutation** — invariant under  $S_6 \times \mathbb{Z}_2$  face relabeling.
3. **Basis changes** — invariant under  $U(n)$  gauge freedom inside degenerate eigenspaces.
4. **Tolerance perturbation** — stable under perturbation of any tolerance within the prescribed regime (§A.1).

Records failing a relevant check remain computational observations or research program items. Current claim status is assigned under the four-level contract in the independent paper, not by this appendix.

## A.6 Gauge Freedoms and Registered Choices

The archived implementation fixes the following representation choices. Alternative choices are admissible when they are declared and cross-validated.

| #   | Freedom                                        | Canonical fixing                                                                                                                                                                                                                         |
|-----|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D.1 | Eigenspace basis ( $U(d_\lambda)$ )            | <code>numpy.linalg.eigh</code> — first nonzero element positive. Real eigenvectors where possible (real symmetric matrix).                                                                                                               |
| D.2 | Sector label ordering                          | <b>CCS canonical:</b> sort by $k = 9(1 - \lambda_{18})$ ascending; within fixed $k$ , sort by dimension ascending. Labels S1–S9 are frozen by Table C3 (§1.4). All figures, tables, and transport tensors <b>MUST</b> use this ordering. |
| D.3 | Sector merging (accidental split)              | Merge sectors whose $(\lambda_{18}, \lambda_{QT}, \lambda_{HT})$ triples differ by $< 10^{-8}$ in all three coordinates. The 11→9 merge pattern is deterministic.                                                                        |
| D.4 | Generator labels ( $S_6 \times \mathbb{Z}_2$ ) | <code>CubieMove.prim_moves</code> enumeration order. Spectral identity (layers, dimensions, projectors) is label-invariant.                                                                                                              |
| D.5 | Isotypic multiplicity ( $GL(m, \mathbb{C})$ )  | Commutant basis from orbit-enumeration construction. Gram-Schmidt orthogonalized.                                                                                                                                                        |
| D.6 | Layer key representation                       | <code>SPECTRAL_DECIMALS = 6</code> . Canonical keys: $\lambda = 1 - k/9$ , $k \in \{0, 1, 2, 3, 4, 6\} \rightarrow [1, 8/9, 7/9, 2/3, 5/9, 1/3]$ .                                                                                       |
| D.7 | Ambient-commutant candidate basis              | Gram-Schmidt orthogonalized conjugacy-class orbit sums. The registered numerical candidate dimension is 610; an exact certificate remains open.                                                                                          |

Projectors, transport strengths, and layer dimensions should agree under valid changes of basis up to the declared matching and numerical tolerances. First-version  $\kappa_d$  diagnostics and the candidate ambient-commutant dimension are retained as archived computational records, not theorem-level gauge invariants established by CCS v2.1.

## Appendix B Provenance

**Purpose.** Record traceable lineage from experiment scripts to structured artifacts, archive tables, figures, and paper-level computational statements.

**Scope.** Data flow diagram, artifact/experiment provenance map, paper usage map, and figure mapping.

**Dependencies.** The paper-local validation and result records indexed below.

**Outputs.** A reviewable provenance map linking retained numerical records to their producing experiments and, where applicable, their consuming papers.

*This part indexes provenance. It does not assign mathematical claim status.*

### B.1 Data Flow

```

experiments and validation scripts -> structured artifacts -> papers
                                   \-> generated figures
                                   \-> CCS v2.1 review index
private raw archive                -> provenance only

```

Executable scripts and structured outputs form the certificate layer. CCS v2.1 indexes selected values and observations for human review. The private raw archive retains failed experiments and older revisions as provenance only.

## B.2 Artifact and Experiment Provenance Map

| Claim                                              | Primary experiment                                                     |
|----------------------------------------------------|------------------------------------------------------------------------|
| 6-layer spectrum, dims, block support              | experiments/paper1/validation/spectral_ladder.py                       |
| registered $k = 5$ vacancy                         | experiments/paper1/validation/k_absence.py                             |
| Projector algebra ( $P_i P_j = \delta_{ij} P_i$ )  | experiments/paper1/validation/projector_algebra.py                     |
| 9 QT/HT joint-spectral sectors                     | experiments/paper2/validation/primitive_sectors.py                     |
| $K$ matrix, transport graph                        | experiments/paper2/validation/transport_graph.py                       |
| Block noncommutativity                             | experiments/paper2/validation/supp_nc.py                               |
| EP algebra census                                  | experiments/paper2/validation/ep_algebra.py                            |
| Graph/operator separation for five canonical paths | experiments/paper3/validation/composition_obstruction.py               |
| Withdrawn spectral-layer $\pi$ map                 | experiments/paper2/archive/commutant_pi_map.py (provenance only)       |
| Withdrawn compressed-commutant census              | experiments/paper1/archive/isotypic_decomposition.py (provenance only) |

## B.3 Subject Navigation

| Subject                     | Primary CCS reference   | Archived material                                    |
|-----------------------------|-------------------------|------------------------------------------------------|
| Averaging-operator spectrum | §§1.1–1.8               | Layers, block reductions, and extended census tables |
| Direct transport            | §§1.4, 2.1–2.2, 2.7–2.8 | Sectors, $K$ matrix, localizer data, and EP algebra  |
| Projected composition       | §2.5                    | Optional copy of the five matrix-obstruction records |

## B.4 Figure Boundary

Figures used by the current archive are placed beside the records they illustrate and are explained by their captions. Historical and withdrawn images remain repository provenance; they are not indexed in this release and do not enlarge the claims of the independent papers.

## Appendix C Figure Provenance

The release PDF includes only figures placed directly in the relevant archive sections. Historical images and presentation-build details are maintained in the repository rather than repeated here.

## Appendix D Implementation Notes

**Purpose.** Document computational methods that underlie the numerical values in Parts I–II. These certify reproducibility without interrupting the mathematical narrative of the main papers.

**Scope.** Representation construction, projector computation, transport and Lie generator algorithms, commutant computation methods, computational complexity table.

**Dependencies.** Part 0.5 (canonical API), Part I (canonical objects), `rime/` source modules.

**Outputs.** Complete algorithmic specification sufficient for independent reimplementa-  
*This part provides the algorithmic specification sufficient for independent reimplementa-  
 tion of the canonical computation.*

Computational methods that underlie the numerical values in Parts I–II. These belong in the supplement, not in the main papers — they certify reproducibility without interrupting the mathematical narrative.

## D.1 Representation Construction

$\rho : G \rightarrow \text{GL}(228, \mathbb{C})$  is built as a permutation+phase representation on corner and edge cubic states. Corner positions are indexed by sign vectors  $\{x \in \{\pm 1\}^3\}$ ; edge positions by vectors in  $\{x \in \{\pm 1, 0\}^3 : \sum_i |x_i| = 2\}$ .

On the **permutation blocks** (CP, EP), generators act by permutation matrices:  $\rho(g)_{ij} = 1$  if position  $j$  maps to position  $i$ , 0 otherwise. These are integer matrices:  $\rho_{\text{cp}}(g) \in M_{64}(\mathbb{Z})$ ,  $\rho_{\text{ep}}(g) \in M_{144}(\mathbb{Z})$ .

On the **orientation blocks** (CO, EO), generators additionally multiply by a phase factor on each affected index:  $\rho(g)_{ij} \in \{0, \pm 1\}$  (EO,  $\mathbb{Z}_2$ ) or  $\rho(g)_{ij} \in \{0, 1, \omega, \omega^2\}$  (CO,  $\mathbb{Z}_3$ , with  $\omega = e^{2\pi i/3}$ ).

**Post- $\rho$ -fix invariant:**  $\|\rho(g)\rho(h) - \rho(gh)\|_F < 3 \times 10^{-8}$  on all blocks — the homomorphism property is exact to machine precision. Verified on 15 random products.

## D.2 Projector Computation

Spectral projectors  $P_\lambda$  are computed via `numpy.linalg.eigh` on  $A_{18}$ :

$$P_\lambda = V_\lambda V_\lambda^H$$

where  $V_\lambda \in \mathbb{C}^{228 \times d_\lambda}$  has orthonormal columns (Schatten normalization).  $\text{Tr}(P_\lambda) = d_\lambda$ ,  $P_\lambda^2 = P_\lambda$ ,  $P_\lambda P_\mu = \delta_{\lambda\mu} P_\lambda$ .

Sector projectors  $P_{S_k}$  are obtained by numerical joint diagonalization of  $A_{18}$ ,  $\text{QT}_{\text{all}}$ , and  $\text{HT}_{\text{all}}$  after the declared commutator audit. Their registered joint clusters group into nine sectors at `CENTER_CLUSTER_TOL` =  $10^{-8}$ . Exact simultaneous spectral projectors require exact commuting-Hermitian registration.

## D.3 Transport and Lie Generators

**Transport:**  $K_{\alpha\beta} = \max_{g \in S} \|P_\alpha \rho(g) P_\beta\|_F$  — enumerates all 18 generators, no optimization needed.

**Historical principal-log registration:**  $A_g = \text{logm}(\rho(g))$  via `scipy.linalg.logm`, using the declared numerical branch. The embedding residual is  $\max_g \|\exp(A_g) - \rho(g)\| < 3 \times 10^{-15}$ . These arrays are retained for first-version reproducibility and are not the anti-Hermitian-part family or a current exact Lie-depth certificate.

**Historical commutator arrays:**  $\kappa_1$  uses  $[A_g, A_h]$  for all 153 unordered generator pairs. The partial  $\kappa_2$  enumeration is exploratory and does not certify Lie closure or saturation.

## D.4 Commutant Computation

**Ambient commutant candidate** (registered dimension 610): a combinatorial orbit-enumeration implementation followed by numerical orthogonalization. The current archive does not contain an independent exact rank/nullity certificate, so 610 remains an unpromoted candidate.

**Per-layer commutant:**

- $d_\lambda \leq 50$ : One-shot SVD on the Kronecker constraint matrix after linear dependency reduction. The constraint is  $CX = 0$  where  $C$  stacks  $G^T \otimes I - I \otimes G$  for each independent generator. Rank threshold:  $\text{tol} \cdot \max(1.0, s_0) \cdot \max(C.\text{shape})$ .
- $d_\lambda > 50$  (only  $V_{5/9}$ ,  $d = 106$ ): Randomized Reynolds iteration — sample budget  $\min(6d, 250)$ , 8 iterations of exact projection, convergence to machine precision.

## D.5 Computational Cost

$N = 228$ ,  $d_\lambda \leq 106$ ,  $|S| = 18$ ,  $K = 3$  (Center operators).

| Operation                               | Complexity                              | Wall time |
|-----------------------------------------|-----------------------------------------|-----------|
| $A_{18}$ eigendecomposition             | $O(d^3)$                                | < 1s      |
| center_decomposition()                  | $O(K \cdot d^3)$                        | < 5s      |
| Ambient commutant candidate             | $O(d^2 \cdot  \text{Conj}(G) )$         | < 1s      |
| Layer commutant ( $d_\lambda \leq 50$ ) | $O(d_\lambda^6)$                        | ~1s each  |
| Layer commutant ( $d_\lambda = 106$ )   | $O(d_\lambda^3 \cdot N_{\text{iter}})$  | ~30s      |
| transport_kappa()                       | $O( S  \cdot d^3)$                      | ~2s       |
| kappa_depth(2)                          | $O(\binom{153}{2} \cdot d^3)$           | ~10s      |
| $\pi$ map SVD (610×966)                 | $O(610 \cdot 966 \cdot \min(610, 966))$ | < 1s      |

Total wall time for full canonical recomputation: ~5–10 minutes on commodity hardware.

## Appendix E Archive Status Register

**Purpose.** Route retained CCS v2.1 material to its current status without restating the independent papers’ theorem spines.

**Scope.** Selected Paper I–II data records, archived extensions, open questions, and withdrawn first-version interpretations.

**Dependencies.** The current Paper I and Paper II ‘Claim Status and Boundary’ sections, declared executable artifacts, and HISTORY.md.

**Outputs.** A compact routing table. It is not a theorem register.

*The independent papers control theorem and certificate wording. CCS v2.1 retains extended human-readable records only.*

| Archive material                                                                     | Current status                                                                                 | Controlling source                               |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Six-layer $A_{18}$ census and block records                                          | Mixed exact derivation, computational certificate, and observation                             | Paper I claim-status section and Paper I scripts |
| Trace-rationality and partition-integrality statements                               | Theorem under the hypotheses stated in Paper I; no canonical-face application is asserted here | Paper I                                          |
| Nine QH sectors, ten direct edges, and EP algebra census                             | Computational certificate                                                                      | Paper II and Paper II scripts                    |
| Generator-family arithmetic and transport extensions                                 | Computational observation                                                                      | CCS v2.1 tables and archived scripts             |
| Candidate ambient-commutant dimension 610                                            | Research program / unpromoted numerical candidate                                              | CCS v2.1 provenance                              |
| Graph/operator composition obstruction                                               | Outside CCS authority; independently certified                                                 | Paper III and its matrix audit                   |
| C0, T7 morphisms, strict word/Lie containment, and old $\kappa$ completion narrative | Historical / withdrawn                                                                         | HISTORY.md                                       |
| Moving spectral/accessibility hierarchies                                            | Research program unless separately certified in the owning paper                               | Papers IV–VII claim boundaries                   |

The repository may retain older tables below in source history, but they are excluded from the current CCS output because their numbering and promotion language predate the v2 paper revisions.

## Appendix F Future Directions and Verification Scope

**Purpose.** Collect CCS-adjacent open problems and computational boundaries. Nothing here is claimed as proven or imported into revised Paper III.

**Verification status.** The current records include the mixed-status six-layer census, nine numerically registered QH sectors, the direct transport matrix, sector non-invariance residuals, and five graph/operator composition obstructions in the 228-dimensional Rubik realization. They do not establish the former T7 strict-containment theorem, a layerwise  $G$ -isotypic decomposition, or an exact 610-dimensional ambient-commutant theorem. Exact statements and finite numerical certificates are separated in Papers I–III.

### F.1 Structural Generalization

**Broader finite-group representations.** CCS experiments include the Rubik cube (228-dim),  $S_3 \text{ nat} \oplus \text{reg}$  (9-dim), and  $S_3 \text{ reg} \oplus \text{reg}$  (12-dim). It remains open whether the Paper II transport architecture generalizes to non-block-diagonal representations, non-symmetric generator families, or infinite discrete groups. The central structural question is which features are  $G$ -determined and which are generator-conditioned. Whether  $\text{Supp}_{\text{nc}}$  extends canonically to non-permutation or non-semisimple transport geometries is unknown.

**Alternative spectral algebras.** The registered Rubik algebra  $\mathcal{B}_{\text{QH}}$  produces nine sectors with a sparse direct transport graph. Alternative commuting algebras may change both the

sector count and the support/composition relation. No refinement-invariance theorem for a compositional gap is currently claimed.

**Graph-to-composition promotion.** The canonical audit instead establishes that a two-step support path need not survive matrix multiplication. A current open problem is to identify transversality, rank, or image–kernel conditions under which graph reachability does imply operator reachability. Comparisons with Lie-generated accessibility require a separate certificate after this operator-level relation is fixed.

**Further structural questions.** Whether the Type I/II classification, the  $M_2$  overlap pattern, or graph/composition obstructions appear in non-permutation representations is open. The candidate ambient-commutant dimension must first receive an independent exact certificate before it is related to transport-graph statistics.

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## F.2 Accessibility Questions

**Algebraic characterization of noncommutative support.** Paper II defines  $\text{Supp}_{\text{nc}}(\alpha)$  from the family-level maximum over the three per-axis QT commutator pairs. In the registered Rubik realization, shared support selects 15 unordered candidates: nine Type I labelled direct edges and six nonedges. It is therefore a localizer, not an edge characterization. Whether a sharper support notion can be derived algebraically from the represented finite-dimensional  $*$ -algebra remains open.

**Promotion conditions.** The former C0–C3 characterization is withdrawn. Candidate replacement hypotheses should act directly on the projected factors, for example through image–kernel transversality, rank protection, singular-value lower bounds, or compatible block support.

**Classification of composition obstructions.** The current five witnesses are explained by physical-block image–kernel mismatch. Whether other systems exhibit incidence, cancellation, or rank-loss obstructions without block separation is open.

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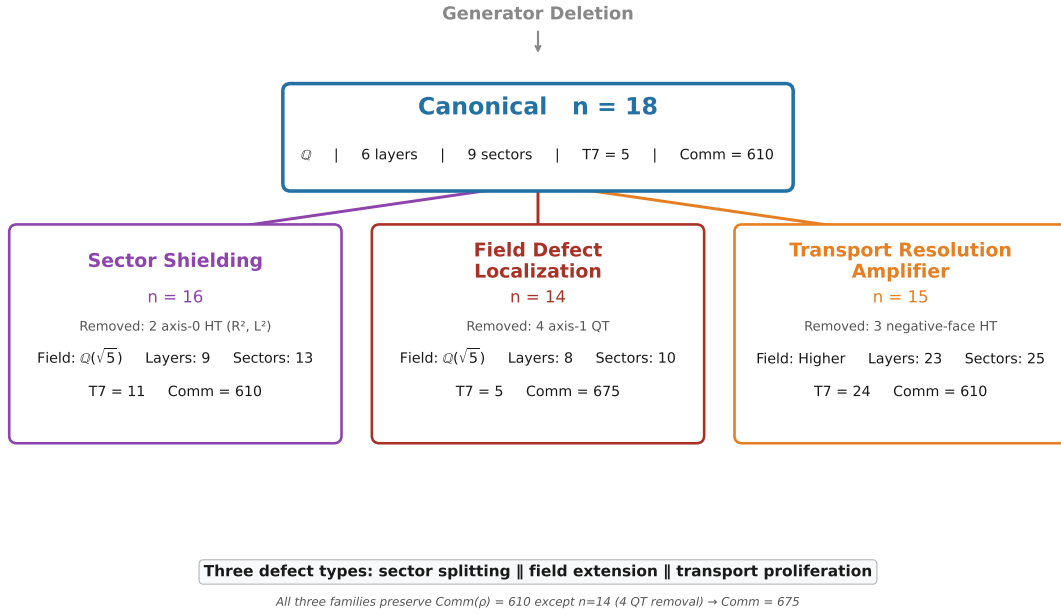
## F.3 Algebraic Extensions

**Generalized transport algebras.** The direct-support norm  $K_{\alpha\beta} = \max_g \|Q_\alpha \rho(g) Q_\beta\|_F$  is defined for the declared sectorization and operator family. What changes when the projectors come from another certified commuting-normal registration? Routed products, full words, commutators, and Lie depth must remain separate typed objects. The first-version  $\kappa$  arrays do not provide the missing bridge.

**Refinement questions beyond  $M_2$ .** The EP census exhibits four registered  $M_2$  components, while the direct graph and localizer record a finite overlap pattern. No theorem states that the presence of  $M_2$  alone forbids every transport-compatible refinement. Identifying the additional hypotheses needed for a refinement obstruction remains open.

**Historical Generator Defect Taxonomy.** Four generator families were constructed by selective deletion from the 18-generator canonical set. The first-version script is archived at `experiments/paper3/archive/generator_defect_taxonomy.py` and is not current matrix-composition evidence.

---



**CCS Figure C18.** Generator defect taxonomy: canonical  $n=18$  and three defect families — Sector Shielding ( $n=16$ ), Field Defect Localization ( $n=14$ ), Transport Resolution Amplifier ( $n=15$ ).

## F.4 Computational Directions

**Scalable commutant extraction.** The current commutant computation uses generator reduction + one-shot SVD ( $d \leq 50$ ) or randomized Reynolds ( $d > 50$ ). For representations beyond  $\sim 1000$  dimensions, both methods become impractical. A scalable commutant algorithm — perhaps exploiting sparse generator structure or block-diagonal preconditioning — is open.

**Automated typed audit.** A future pipeline may take a declared representation, sectorization, operator family, dtype, and tolerance policy and emit separate direct-support, routed-product, full-word, commutator, and Lie-closure records. It must preserve these types rather than reconstruct the withdrawn  $K \rightarrow \kappa_0 \rightarrow \kappa_1 \rightarrow T7$  ladder.

**Exact QH-algebra reconstruction.** The commuting algebra  $\mathcal{B}_{QH} = \text{alg}(A, QT_{\text{all}}, HT_{\text{all}})$  is presently supported by numerical commutation and joint-sector certificates. An exact reconstruction would promote this numerical object without identifying it with the center or the full ambient commutant.

**Separate Lie-accessibility audit.** First-version  $\kappa_d$  calculations are archived diagnostics. Any future comparison between Lie accessibility and projected word composition requires a new claim-specific certificate.

## F.5 Structural Scope Boundary

What the current CCS does NOT claim:



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| Claim                                                                    | Status                                                                                                            |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| A support-graph path implies nonzero projected composition               | <b>Refuted in general.</b> The five canonical paths are obstruction witnesses.                                    |
| The CCS applies to AGI, cognition, planning, solver algorithms, robotics | <b>Not claimed.</b> Its scope is finite-group representation computation.                                         |
| The CCS is a general classification of finite-group representations      | <b>Not claimed.</b> Rubik is one finite computational realization.                                                |
| The first-version $\kappa_d$ /T7 hierarchy proves a composition gap      | <b>Withdrawn.</b> A separate operator-level certificate is required.                                              |
| The registered direct graph is directed or asymmetric                    | <b>Not supported.</b> The inverse-closed family gives a symmetric $K$ matrix to the declared numerical tolerance. |

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**Code availability.** Code and computational certificates are available in the [RIME repository](#).

*End of the CCS v2.1 computational companion archive. Independent papers and declared executable artifacts control current claims and certificates; withdrawn source sections are retained for provenance only.*